



OPEN A new intrusion detection method using ensemble classification and feature selection

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Intrusion Detection Systems (IDS) play a crucial role in ensuring network security by identifying and mitigating cyber threats. This study introduces a hybrid intrusion detection approach that integrates Convolutional Neural Networks (CNNs) for feature extraction and the Random Forest (RF) algorithm for classification. The proposed method enhances detection accuracy by leveraging CNNs to automatically extract relevant network features, reducing data dimensionality and noise. Subsequently, the RF classifier processes these optimized features to achieve robust and precise intrusion classification. To evaluate the effectiveness of the approach, experiments were conducted on the KDD99 and UNSW-NB15 datasets. The results demonstrate that the proposed model achieves an accuracy of 97% and a precision of over 98%, outperforming traditional machine learning-based IDS solutions. These findings highlight the potential of the proposed hybrid framework as a scalable and efficient cybersecurity solution for real-world network environments.

Keywords Intrusion detection system (IDS), Machine learning, Neural networks, Random forest

With the increasing prevalence of computer networks, the security of these networks has become a top priority¹. Every computer is always at risk for unauthorized and intrusion, however, with sensitive and private information are at a higher risk^{2,3}. Numerous machine learning methods, including nature-inspired metaheuristics, Bayesian algorithms, and swarm intelligence, have been previously proposed for feature selection and IDS performance improvement⁴. Intrusion Detection is a key technique in Information Security plays an important role detecting different types of attacks and secures the network system⁵. Nevertheless, security and privacy vulnerabilities in CPS-IoT networks remain key issues, as anomalies may lead to significant and system-wide consequences⁶. Intrusion Detection is the process of observing and analyzing the events arising in a computer or network system to identify all security problems^{7–9}. Intrusion detection systems are as one of main factors of security sub-structures in most organizations¹⁰. These systems are hardware and software models and patterns which consider automating processes which these processes investigate to monitor events in network or computer systems. The importance of optimizing resource distribution and reducing latency in distributed systems is related to the aim of improving the accuracy and speed of intrusion detection systems through the combination of neural networks and centralized machine learning algorithms. This issue indicates the importance of using intelligent algorithms to improve the performance of complex systems¹¹. Novel models for identifying threat intrusions in IIoT are executed via a genetic algorithm incorporating an attention mechanism and an optimized LSTM utilizing Adam (GA-mADAM-IIoT)¹². Intrusion detection systems survey and analyze these events for problem solving of security of systems and computer networks^{13,14}. CNN models extract features from varied datasets for speaker recognition efficiently, supporting its usage in our proposed intrusion detection system for feature selection from network traffic data. Their comparison with LSTM models shows that CNNs can improve classification accuracy, supporting our CNN-RF method¹⁵. Although most of these warnings are not correct and they are so-called wrong warnings, but there is no evidence to show that these warnings are correct or not which this issue needs too much survey¹⁶. A cost-effective coordination method for low-latency applications in NFV networks has been implemented with a hybrid CNN-RF intrusion detection approach, where efficient feature selection by CNNs reduces computational overhead and enhances real-time detection performance in network environments¹⁷. In intrusion detection systems, full mesh aggregation to coordinate the service operation chain across multiple domains can improve network efficiency. This concept is supported by emphasizing the value of optimal data

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processing, similar to how CNNs in the proposed model of this research modify features to increase the accuracy of Random Forest intrusion detection¹⁸. investigated the live migration of correlated virtual machines in cloud data centers for the CNN-RF model and how to reduce feature dimensions to improve the efficiency of intrusion detection in network systems, highlighting efficient resource management¹⁹. By integrating real-time multi-source data, CNN features are extracted from network traffic, while meta-learning optimization enhances the efficiency of the neural network and supports the role of CNN in handling complex anomalies. Noise-resilient feature extraction can reflect the proposed approach, where CNNs refine noisy network data for accurate RF classification, and overall enhance the detection performance^{20–22}.

Deception-resistant network systems use CNNs to extract important information from possibly corrupted network traffic, allowing RF to identify intrusions despite hostile interference. It also proposes an effective prediction-based technique for mobile population sensing that optimizes data selection, reduces CNN feature size to improve RF classifier efficiency, and prioritizes key network data features for intrusion detection^{23,24}. The CNN-RF hybrid intrusion detection system filters out irrelevant or noisy network features and improves the RF classifier's accuracy and robustness against low-quality data inputs. The federated non-learning approach increases model reliability in distributed systems²⁵. Data storage and retrieval optimization help CNN reduce high-dimensional network data into concise and relevant feature sets to improve RF intrusion classification²⁶.

The CNN-RF intrusion detection system uses fast and lightweight similarity detection to compress data and reduce redundancy. CNNs remove redundant network features to help the RF classifier process data faster and improve detection²⁷. However, To detect multimodal fake news, the cognitive-based inference network used CNNs to extract coherent and meaningful information from varied network traffic²⁸. Increase model robustness by filtering harmful or noisy network input with CNN to prevent poisoning attacks in IIoT federated learning and ensuring the RF classifier offers accurate intrusion detection results²⁹. Multi-modal fusion for fake news identification, where CNNs aggregate and modify complicated network traffic aspects to help RF classify intrusions by making sense of conflicting data³⁰. Hierarchy and class imbalance in network intrusion detection that improves this CNN-RF model, where CNNs handle complex feature hierarchies and imbalances in network data. Moreover, these systems may struggle to adapt to evolving user behavior or system changes, which can lead to reduced accuracy over time³¹.

Figure 1 illustrates the concept of how SIDS functions. In SIDS, the primary approach involves creating a database of intrusion signatures or patterns, which represent known malicious activities or attacks³². Influential nodes in social networks are identified using advanced step-wise evaluation search, improving the efficiency of network analysis that supports the CNN-RF model, in which CNNs extract key features from network traffic, similar to identifying critical nodes, and enhancing the ability of RF to accurately detect intrusions³³. Federated learning incentive mechanism with smart contracts in AIoT that ensures secure and efficient data collaboration to achieve reliable intrusion detection in distributed systems³⁴. Sustainable online training platforms analyzed using big data and optimized on scalable data processing to optimize performance on large-scale network data, and increasing the efficiency of RF classification and intrusion detection accuracy³⁵. High-accuracy intrusion classification using supervised kernel PCA and kriging for high-dimensional surrogate modeling is developed to reduce the features of CNNs in network data to optimize the performance of complex systems³⁶. In general, this means that if the network pattern is equal to the conditions in the intrusion detection system database, it means that there is a possibility of intrusion, and if it is not the same as any of the patterns defined in the IDS database, it means that the traffic is normal.

Fast GNSS acquisition algorithm using SFFT for RF ability to effectively detect intrusion patterns that enhance signal detection in noisy environments³⁷. This enables CNNs to filter out network traffic noise. Such anomalies are considered potential intrusions, as they indicate deviations from expected behavior. This flexibility allows AIDS to detect novel or zero-day attacks, which are previously unknown and not covered by predefined signatures in SIDS. Table 1 provides a comparison between signature-based detection and anomaly-based detection the proposed model aims to enhance the accuracy and efficiency of intrusion detection systems

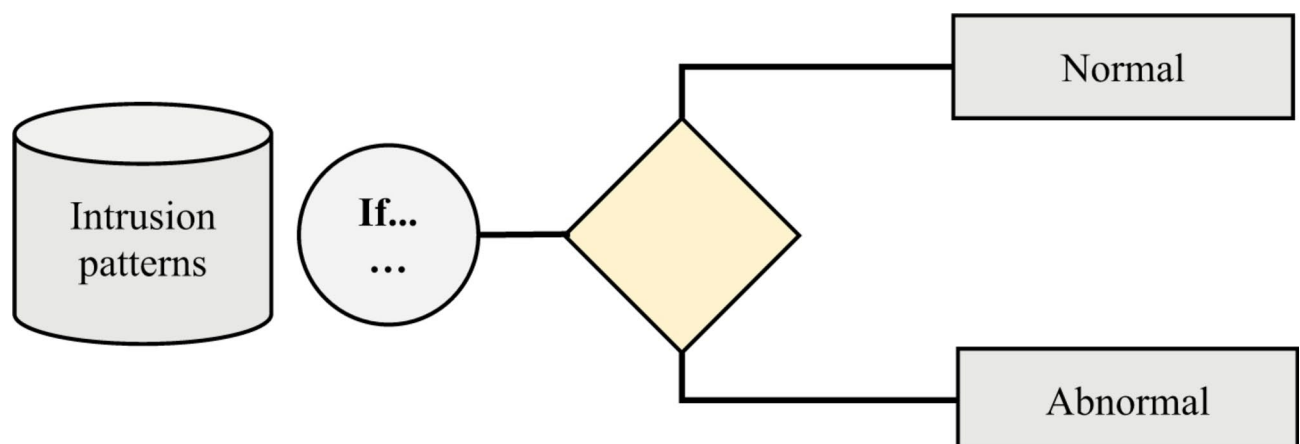


Fig. 1. IDS concepts.

Criteria	Signature-Based Detection	Anomaly-Based Detection	Behavior-Based Detection	Hybrid Detection Systems
Detection Approach	Matches against predefined intrusion signatures.	Compares against a model of normal behavior.	Analyzes behavioral patterns for deviations	Integrates multiple detection methods for comprehensive analysis
			Effective against both known and unknown threats	Offers adaptability to evolving threats
Known vs. Unknown Attacks	Detects only known intrusions.	Can identify unknown or zero-day attacks.	Can adapt to changing attack vectors through behavioral analysis	Combines signature-based and anomaly-based approaches for enhanced adaptability
			Can be tuned to minimize false positives, but may still have moderate false positive rates	Strives to minimize false positives through comprehensive analysis of multiple factors
Adaptability to New Threats	May struggle with new and evolving attack patterns.	More adaptable to new and emerging threats.	Can overlook anomalies if behavioral patterns are not accurately captured	Aims to reduce false negatives by integrating diverse detection methods
			Scalable, but computational overhead may increase with complex behavioral models	Balances scalability with performance to accommodate varying network sizes
False Positive Rate	Typically, lower false positive rate due to specific signatures.	May have a higher false positive rate due to normal activities being flagged.	Can be resource-intensive, particularly for real-time analysis of behavioral data	Strives to optimize resource utilization through efficient algorithmic design
			Scalable, but computational overhead may increase with complex behavioral models	Offers tailored solutions for securing IoT networks, leveraging hybrid approaches for comprehensive protection
			Can be resource-intensive, particularly for real-time analysis of behavioral data	Provides tailored detection mechanisms for evolving IoT threat landscape
Use of Machine Learning	Less reliance on machine learning techniques.	Often employs machine learning for anomaly detection modeling.	Well-suited for IoT environments, providing granular insight into device behavior	provides specialized IoT network security solutions by utilizing hybrid ways for all-encompassing defense.
			Well-adapted to detecting IoT-specific threats through behavioral analysis	combines performance and scalability to handle different network sizes.
Scalability	Generally scalable, but may require frequent updates to signature databases	Scalable, but performance may degrade with large-scale deployments	Scalable, but computational overhead may increase with complex behavioral models	Balances scalability with performance to accommodate varying network sizes
Resource Requirements	Moderate resource requirements for signature matching	May require substantial computational resources for anomaly modeling	Can be resource-intensive, particularly for real-time analysis of behavioral data	Strives to optimize resource utilization through efficient algorithmic design
Integration with IoT Devices	Can be integrated with IoT devices, but may require specific signatures for IoT-related threats	Compatible with IoT environments, but may require customized anomaly models for IoT traffic	Well-suited for IoT environments, providing granular insight into device behavior	Offers tailored solutions for securing IoT networks, leveraging hybrid approaches for comprehensive protection

Table 1. Differences between signature-based detection and anomaly-based detection.

(IDS) by integrating feature selection and ensemble classification techniques. The objectives include improving the rate of false positive and false negative warnings, increasing overall accuracy, enhancing detection efficiency, and identifying relevant features for intrusion detection. By utilizing conventional neural networks (CNN) for feature extraction and RF for classification, the model aims to achieve an effective pattern in intrusion detection with high detection rates and reduced false alarm rates. Evaluation using the KDD99 dataset demonstrates superior performance metrics, with the proposed solution achieving a remarkable accuracy rate of 97% and precision exceeding 98%, surpassing traditional algorithms and showcasing its effectiveness in mitigating network intrusions. The author contribution of this article are as follows:

- Improving the rate of false positive and false negative warnings in intrusion detection.
- Increasing the overall accuracy of intrusion detection systems.
- Enhancing the efficiency of intrusion detection processes.
- Identifying relevant and appropriate features for intrusion detection.
- Achieving an effective pattern in intrusion detection systems with high detection rates and reduced false alarm rates.

The proposed research introduces a hybrid intrusion detection solution that utilizes conventional neural networks as the foundational function along with the RF algorithm. This approach aims to offer an advanced and effective solution for intrusion detection. The rest of the paper is organized as follows. The research literature is reviewed in Sect. 2. In Sect. 3, the approach of the proposed model and its analysis are described in detail. The evaluation and implementation of the proposed method is given in Sect. 4. Finally, in Sect. 5, conclusions and future works are discussed.

Related works

Intrusion detection research has witnessed extensive exploration and development of techniques and models to enhance accuracy and effectiveness. The primary goal is to create robust methods capable of detecting unauthorized access, malicious activities, and network intrusions. Adaptive data processing, integrated excitation and sensing for intelligent soft robots in intrusion detection scenarios, where CNNs adaptively extract relevant network features³⁸. These methods involve training models on labeled datasets to learn normal and

anomalous behavior patterns, enabling them to identify deviations in network traffic. Maximum throughput in secure multipath transmission for ad-hoc wireless networks, focusing on efficient data management, where CNN-based feature selection optimizes network data processing and increases the efficiency and accuracy of RF intrusion detection³⁹.

The researchers in⁴⁰ introduce a novel hybrid autoencoder and modified particle swarm optimization (HAEMPSO) for feature selection, alongside a deep neural network (DNN) for classification. Particle Swarm Optimization with inertia weight adjustment was employed to optimize the parameters of the Deep Neural Network. An experimental investigation was performed on two authentic datasets, UNSW-NB15 and BoT-IoT, which are appropriate for the IoT scenario.

The paper⁴¹ introduces a unique ensemble-based approach to increase intrusion detection in the context of modern 5G embedded and cyber-physical network security. The suggested method fortifies modern networks' defenses against possible intrusions by utilizing an optimized CatBoost classifier. The KDDCup99 dataset was used in studies to assess the efficacy of the suggested strategy. The findings of the suggested approach demonstrate an impressive 99.96% intrusion detection accuracy. Additionally, several IoT network devices are installed on-site without hardware security protocols. This indicates that conventional intrusion detection systems (IDS) are inadequate for safeguarding the IoT network ecosystem. To resolve these challenges, research⁴² introduces the IoT-Defender framework, which integrates a modified genetic algorithm (MGA) model with a deep short-term memory (LSTM) network for the detection of cyberattacks in IoT networks.

The output accuracy is one of the documented solutions that have been created and/or investigated on intrusion detection techniques between 2000 and 2019, according to the study⁴³. A thorough overview of the different papers will be given with the aid of this survey⁴⁴. Critical density for coverage in camera sensor networks under irregular obstacles where CNNs extract essential features from complex network data and enhance the accuracy of RF intrusion detection in challenging environments.

CNNs filter out unique network patterns, similar to vibration signals, and enhance the accuracy of RF in detecting intrusions. Thus⁴⁵, developed a two-factor authentication system using finger vibrations and improved security in mobile phone systems. Research⁴⁶ studied the trade-offs of code estimation errors for SCER attacks and emphasized robust signal processing where CNNs reduce noise and errors in network data and enable RF to classify intrusions with higher reliability. A flexible authentication scheme called CALRA for vehicle congestion sensing is proposed in⁴⁷, which ensures secure data exchange where CNN-based feature selection secures network traffic analysis and enhances RF performance in intrusion detection. Researchers in⁴⁸ introduced an unsupervised federated hypernetwork for multivariate time series anomaly detection. This supports the CNN-RF system, as CNNs process temporal network data. In⁴⁹, a profile-based predictive handover strategy for 5G networks, optimizing network efficiency and latency, called PBPHS, is proposed, which is consistent with the CNN-RF approach, where CNNs extract critical network features to reduce latency and enhance the performance of real-time RF intrusion detection in high-speed environments.

A low computational consensus control for nonlinear multi-agent systems with sensor errors is proposed in⁵⁰ with an emphasis on efficiency, where CNNs reduce the computational burden by extracting key features. In⁵¹, a reinforcement learning approach is introduced for nonlinear system control using event-driven dynamic mechanisms that adaptively modify network data and improve the RF's ability to detect intrusions in dynamic and complex environments. The study in⁵² developed an event-driven optimal control for stochastic nonlinear systems with a focus on disturbance management, where CNNs filter network traffic disturbances and increase the RF's accuracy in classifying random intrusion patterns. The experiments conducted on different datasets provide valuable insights into the effectiveness and performance of the proposed methods in detecting intrusions in computer networks.

Research⁵³ tackles the increasing difficulty of identifying zero-day botnet assaults within the Internet of Vehicles (IoV), a sophisticated network that facilitates real-time communication between vehicles and adjacent infrastructure to better traffic management, safety, and the driving experience. Conventional machine learning-based intrusion detection systems (IDS) for the Internet of Vehicles (IoV) encounter two primary constraints: the necessity for extensive labeled datasets and the "black box" phenomenon, wherein the rationale behind the model's conclusions remains opaque, hence diminishing confidence among users and stakeholders.

The goal of this method is to classify normal and abnormal data with high accuracy and reduce the error rate. In this paper, the combination of SVM with a self-organizing ant colony clustering network is presented. The proposed method is evaluated using the KDD99Cup dataset to assess its effectiveness in intrusion detection., and the results indicate the superiority of the combined approach over each individual method of SVM and self-organizing ant colony clustering network. Table 2 presents a comparison of the existing methods in the research background.

Several recent studies have explored various machine learning and deep learning techniques for intrusion detection, including ensemble models, feature selection strategies, and hybrid frameworks. For example, methods such as HAEMPSO-DNN and IoT-Defender have integrated autoencoders and LSTM networks for anomaly detection, while ensemble approaches like CatBoost and hybrid optimization techniques have shown promising results in intrusion classification. However, many of these studies either lack an efficient feature extraction mechanism or suffer from high computational overhead, making them less suitable for real-time applications. The proposed research distinguishes itself by utilizing CNNs for automated feature selection, which significantly reduces data dimensionality and eliminates irrelevant attributes, leading to faster and more accurate classification. Additionally, the combination of CNNs with the RF classifier enhances the model's robustness against noisy data, outperforming conventional deep learning-based IDS solutions. The evaluation results on KDD99 and UNSW-NB15 datasets confirm that this approach achieves higher accuracy and precision compared to traditional and state-of-the-art methods, making it a scalable and efficient solution for modern network security challenges.

Ref.	Dataset	Learning method	method	IDS Type	Security Measures (Machine Learning)	Security Measures (Intrusion Detection)
7	ADFA-LD	Supervised	Multilayer perceptron neural network	Signature based	Feature selection to enhance model generalization	Integration of multiple anomaly detection algorithms for robustness
18	KDD99Cup	Semi-supervisory	Deep neural network	Signature based	Data augmentation to address class imbalance	Ensemble learning techniques for improved detection performance
54	Simulated Network	Supervised	Apollonius circles-based handover	Anomaly based	Geometric optimization for efficient data transitions	Monitoring network anomalies during handover in cellular networks
55	Wi-Fi Traffic	Unsupervised	WiShield privacy framework	Anomaly based	Signal processing to anonymize Wi-Fi data	Detecting privacy intrusions via Wi-Fi tracking anomalies
56	UNSW-NB15	Supervised	Func-Bagging ensemble	Anomaly based	Ensemble bagging for robust anomaly modeling	Enhanced detection of heterogeneous network intrusions
57	CICIDS2017	Supervised	Explainable traffic detection	Anomaly based	Interpretable feature extraction for traffic analysis	Explainable identification of malicious network traffic
60	NSL-KDD	Reinforcement learning	Fuzzy weight-based control	Anomaly based	RL-based fuzzy weighting for multi-agent security	Secure anomaly detection in heterogeneous network systems

Table 2. An overview of the existing methods.

Despite significant advancements in IDS using machine learning and deep learning techniques, several research gaps remain. Many existing models, such as deep neural networks (DNNs) and ensemble-based approaches, suffer from high computational complexity, making them less suitable for real-time applications. Additionally, feature selection methods in prior works often fail to effectively reduce data dimensionality while retaining critical information, leading to increased processing time and lower accuracy. Furthermore, traditional IDS models struggle with class imbalance in datasets like KDD99, which can bias classification results. To address these challenges, this research proposes a CNN-RF hybrid model that enhances feature extraction efficiency, improves classification accuracy, and mitigates class imbalance, offering a scalable and computationally efficient solution for modern intrusion detection.

Method

In this section, details of the proposed hybrid method are presented. To implement the proposed solution, the suggested model in Fig. 2 is utilized in the networks. The following provides an explanation of the details of each section. In the proposed method, before applying the data to the suggested model, a preprocessing step is performed to normalize the data. This section includes the following stages:

- Data Homogenization (Training and Testing Data): In this stage, homogenization is applied to the data at the data level by replacing alphabetical characters in the dataset with numerical values of a specific type.
- Data Standardization (Training and Testing Data): In this stage, the data imbalance is addressed. Since the KDDcup99 dataset is used, some features have large numerical values that can dominate other features. In this section, this type of anomaly is eliminated.

In the next module, the operations related to the neural network are performed. Specifically, by utilizing conventional neural networks (CNN), important features of the data are extracted to enhance the overall performance of the solution and the efficiency of the RF algorithm. Feature reduction can ultimately improve the efficiency of the RF algorithm. In methods that lack this capability, the learning algorithm is forced to use features that have no specific relevance to intrusion and, in other words, do not play a role in determining attacks. This type of learning essentially involves learning with noisy data and has a negative impact on the intrusion detection solution.

In the proposed solution, the RF algorithm is employed for learning; however, it deviates from the conventional approach by using only the important features provided by the neural network component. This selective utilization of features leads to improved performance in the RF algorithm. By focusing solely on relevant features, the processing, deployment, and utilization of irrelevant features that do not significantly contribute to classification are avoided. This results in enhanced intrusion detection and faster learning speed. The utilization of conventional neural networks in this context is driven by several main reasons:

- Dimensionality Reduction: Conventional neural networks facilitate a mapping from the raw data space to a conceptual space. This process involves reducing the data’s dimensionality and selecting essential features. As a result, the neural network component plays a crucial role in extracting relevant information from the data.
- Feature Selection: By employing conventional neural networks, the model can automatically identify and prioritize the most important features for classification. This inherent feature selection capability allows the RF algorithm to focus on meaningful data attributes, leading to more accurate results.
- Non-linear Mapping: Conventional neural networks are well-suited for capturing non-linear relationships between features in the data. In complex intrusion detection scenarios, where traditional linear methods may fall short, neural networks can effectively model intricate patterns and correlations.
- Generalization: Conventional neural networks possess strong generalization capabilities. This means that once trained on a sufficiently diverse dataset, they can effectively classify unseen instances and adapt to new intrusion patterns.

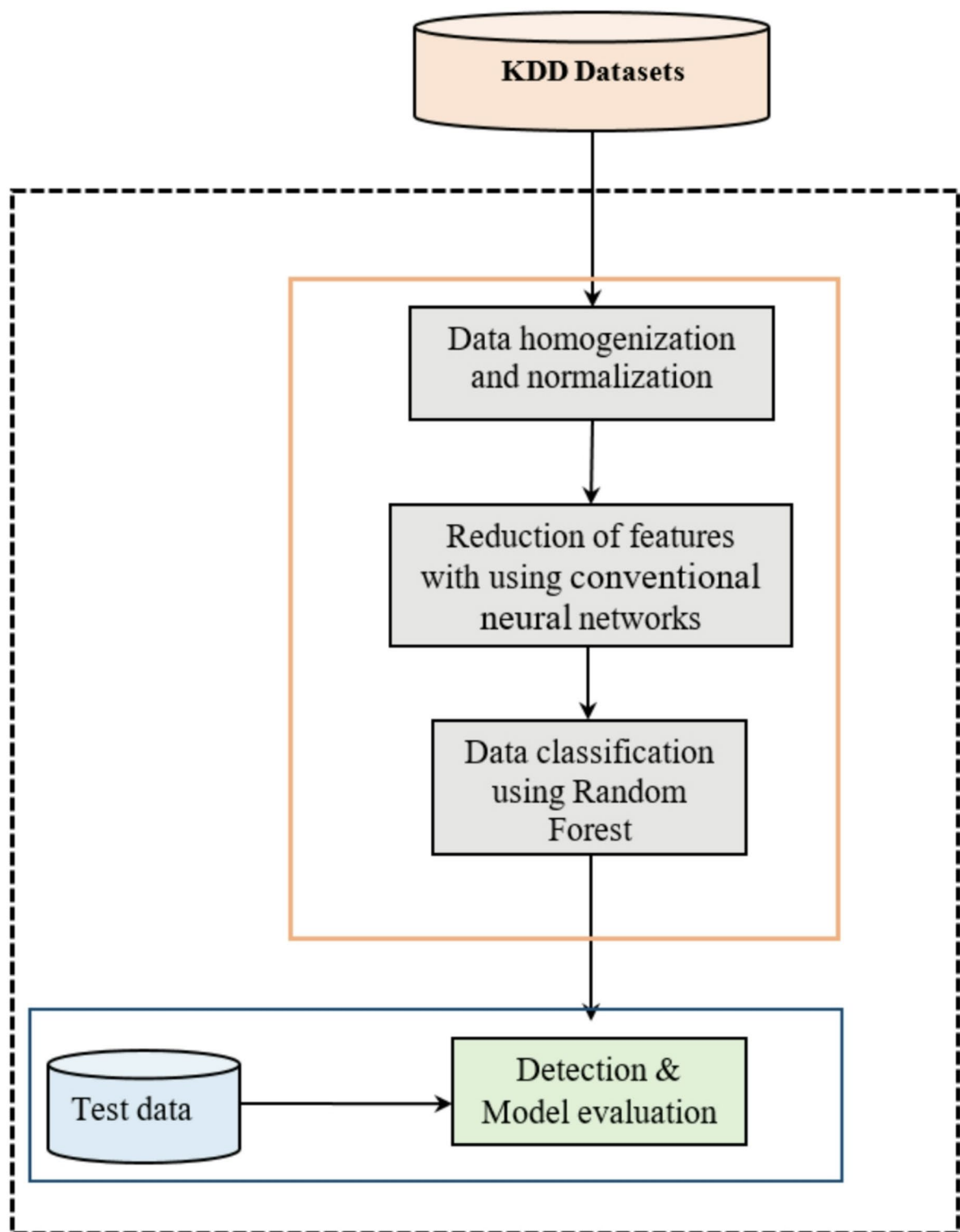


Fig. 2. Proposed method.

- **Learning Representations:** Conventional neural networks learn hierarchical representations of the data, which can be valuable for understanding the underlying structure of the intrusion detection problem. These learned representations enable the model to capture intricate details and nuances in the data.

Overall, the combination of conventional neural networks and the random forest algorithm in the proposed hybrid method leverages the strengths of both approaches, resulting in an efficient and effective intrusion

detection solution that benefits from feature reduction, improved performance, and enhanced generalization capabilities.

Feature reduction

After preprocessing the data, the next stage involves the utilization of conventional neural networks (CNN) to perform the learning process on the preprocessed dataset and subsequently reduce the number of features^{58,59}. CNN neural networks are considered one of the most powerful types of neural networks, particularly in function approximation problems. They surpass perceptron neural networks and excel in various areas, such as learning, identification, similarity analysis, and control of dynamic nonlinear systems. The neural network's feature selection process leverages its inherent ability to capture complex, nonlinear relationships among input features. By passing data through layers of neurons, the network learns hierarchical representations, where irrelevant or redundant features are naturally suppressed. For instance, in a convolutional neural network (CNN), the feature maps generated in initial layers are refined through pooling and activation functions, effectively filtering out noise and highlighting critical patterns. This systematic extraction of essential features ensures that only the most informative and discriminative characteristics of the data are retained for subsequent processing.

When these refined features are input to the random forest algorithm, its performance is significantly enhanced. Since random forest relies on decision trees that partition the feature space, providing it with optimized input reduces ambiguity and overfitting. This synergy allows the random forest to build more accurate splits, leading to better generalization and improved prediction accuracy. Additionally, the dimensionality reduction achieved through neural network preprocessing decreases the computational complexity for the random forest, further enhancing its efficiency in handling large datasets.

Figure 3 visually depicts how these networks are leveraged for feature reduction and intrusion detection. In this stage, the CNN neural networks take the preprocessed data as input and employ their advanced capabilities to learn and extract essential features from the dataset. By doing so, the CNNs significantly reduce the dimensionality of the data and retain only the most pertinent information for intrusion detection. The feature reduction achieved by CNNs is of paramount importance for the overall performance of the intrusion detection system. By focusing on the most relevant features, the subsequent steps of the proposed hybrid method, particularly the implementation of the RF algorithm, become more efficient and accurate. The reduced feature set allows the RF algorithm to concentrate its efforts on crucial attributes, leading to improved classification and detection of intrusion instances.

The class imbalance issue in the KDD99 dataset is effectively handled using a data preprocessing step. In particular, in the Data Standardization section, normalization techniques are used to reduce the influence of larger numerical values that can dominate other features and to prevent the model from biasing towards the dominant class (generally normal traffic). In addition, the CNNs employed in the proposed model indirectly help balance between classes by extracting relevant features and reducing the dimensionality of the data, since focusing on key features reduces the dependence on the number of samples in each class. The RF algorithm is also robust to class imbalances due to its combinatorial nature and the use of majority voting, preventing bias in favor of the majority class.

In the proposed approach, the Convolutional Neural Network (CNN) is responsible for extracting the most relevant features from the original dataset, reducing the dimensionality of the data while preserving critical information. The KDD99 dataset, which originally contains 41 features, was processed using the CNN-based feature selection mechanism, reducing the feature set to 18 highly informative features. Similarly, for the UNSW-NB15 dataset, which initially includes 49 features, the CNN feature extraction process selected 22 key features that contribute significantly to classification performance. This reduction minimizes redundant and irrelevant attributes, leading to improved classification accuracy and reduced computational complexity for the RF classifier.

Figure 3 provides a visual representation of how this process unfolds, showcasing the key role of CNNs in feature extraction and reduction for enhanced intrusion detection.

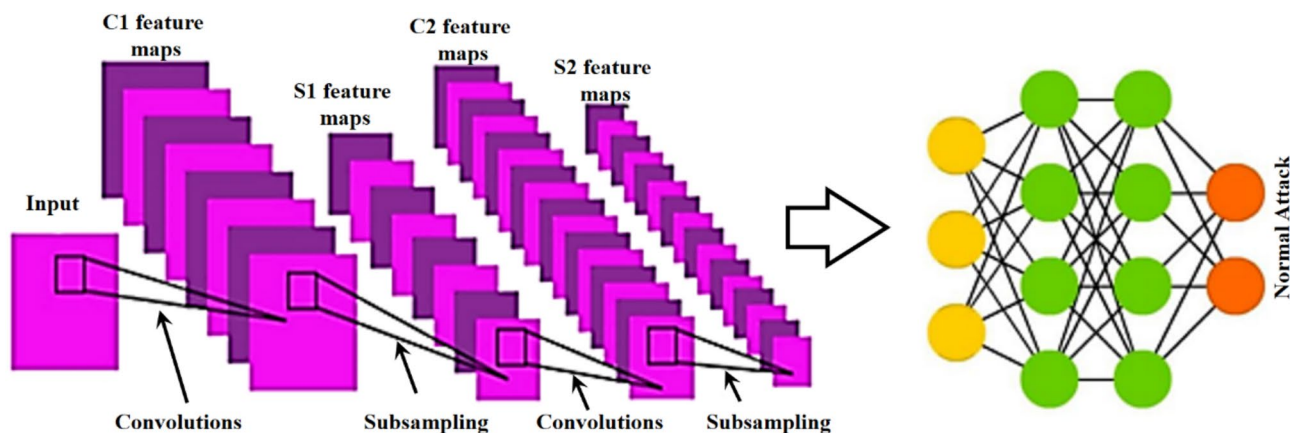


Fig. 3. CNN structure for IDS.

In the conventional neural networks, each layer contains a distinct weight matrix, bias vector, input vector, and output vector. The network can have varying numbers of neurons in different layers, with the output layer being responsible for producing the final network output. All layers between the input and output layers are considered hidden layers⁶¹. Using multiple layers in a neural network increases its efficiency compared to single-layer networks. A two-layer network with a first sigmoid layer and a second linear layer can approximate most arbitrary functions and is commonly used. Pupil morphology is used for personalized recommendations in smartphones and enhances pattern recognition in data⁶². In the context of this specific problem, the neural network is designed to calculate the F0 value for each signal, resulting in an output neuron per signal. Detection of fraudulent taxi trips using route data, with a focus on anomaly detection in the Internet of Things, has been done in⁶³ because CNNs detect abnormal network behavior. If there are too few neurons, the network might lack the necessary resources to handle complex nonlinear situations effectively⁶⁴. On the other hand, an excessive number of neurons can lead to longer training periods and the risk of the network overfitting the training data, making it less adaptable to real-world problems. To address this, trial and error are commonly used to determine the appropriate neuron count in the hidden layer. In the problem being discussed, based on the best experimental findings for the intended network, a hidden layer with 13 neurons has been chosen. In general, the neural network architecture, which enhances feature extraction from complex data, including the number of layers and neurons, plays a significant role in the network's ability to accurately approximate F0 values for each signal and effectively detect intrusions in the proposed hybrid method⁶⁵. The following formula is used to determine the output of the hidden layer:

$$OP = SGM \left(\sum x_i w_{tm}^h \right) \quad (1)$$

The low-rank approximation and kernel learning method for microRNA-disease association, optimizing the data representation, when the weight vector is doubled, the output of the hidden layer is transferred to the output layer⁶⁶. This can be illustrated as follows, considering the output of the i th neuron in the bottom layer:

$$O_i = Sgm \left(\sum_m Sgm \left(\sum x_i w_{tm}^h \right) w_{mi}^o \right) \quad (2)$$

Graph autoencoder for inferring gene regulatory networks, focusing on modeling relational data where h and O denote the hidden and output layers, respectively, and w the layer weights. Sgm is also a sigmoid function, defined as follows⁶⁷:

$$Sgm = \frac{1}{1 + e^{-x}} \quad (3)$$

In neural networks, learning involves adjusting the synaptic weights to align the neuron's input-output relationship with a predefined objective. Various approaches have been employed for training neural networks, and one of these techniques is the error backpropagation algorithm. To evaluate the network error, a comparison is made between the network's output and the desired output. To minimize this discrepancy, the network's output is calculated at each step, and the weights are adjusted based on the difference from the expected output. The error is minimized by leveraging the simplicity and widespread use of the gradient descent method to compute the squared error between the network's outputs and the target function.

The error backpropagation approach is the most widely used method in contemporary neural network applications, accounting for approximately 95% of implementations. A secure reinforcement learning method for mobile network resource allocation, which ensures robust performance. This approach repeatedly adjusts synaptic weights layer by layer to gradually reduce the prediction error. It detects errors in nodes of previous and subsequent layers⁶⁸. The error in this scenario is described as follows:

$$E(\vec{w}) = \frac{1}{2} \sum_{d \in D} \sum_{k \in \text{outputs}} (t_{kd} - o_{kd})^2 \quad (4)$$

The term "Outputs" in this context refers to the target values, specifically the output corresponding to the k th output unit in the training example d , denoted as t_{kd} and o_{kd} for the set of units from the output layer. It is important to note that the slope gradient approach aims to minimize the error to reach an appropriate hypothesis. Deep learning graphs and urban knowledge are used to generate cellular traffic that can improve data scalability and also improve the performance of device-to-device communication in wireless networks by focusing on efficient data exchange, where CNNs simplify the processing of network features and enhance RF intrusion detection in distributed systems. However, there is no guarantee that this algorithm will always converge to the absolute minimum value. These are the weights of the connections from the input layer to the hidden layer and from the hidden layer to the output layer, respectively^{69,70}. The error for the output layer is calculated as follows:

$$\delta_k = O_k(1 - O_k)(t_k - O_k) \quad (5)$$

The error is identified as follows:

$$\delta_k = O_k(1 - O_k) \sum_k w_{kh} \delta_k \quad (6)$$

Each weight is changed as follows:

$$w_{ji} = w_{ji} + \Delta w_{ji} \quad (7)$$

Where:

$$\Delta w_{ji} = \eta \delta_j + X_{ji} \quad (8)$$

The weights of the neural network are adjusted repeatedly during the training phase to attain the optimal weights for the solution. After the training stage, the network is presented with test data that it has never encountered before, known as the test stage. Based on the learned weights from the training phase, the network's performance, often represented as F_0 , is evaluated. During the learning phase, the network receives input data one by one. This input data is processed through the network, where it is multiplied by the synaptic weights, and the resulting output from each neuron serves as input for the neurons in the subsequent layer. This information flow continues until the output layer produces a response from the network. Since random synaptic weights are initially used, the calculated response of the network will differ from the experimentally measured output, leading to an error. The objective is to minimize this error, making the difference between the network's anticipated response and the experimentally measured output as close to zero as possible to achieve accurate predictions⁷¹.

In each iteration, when a new sample enters the network, the process of error backpropagation is applied once again to adjust the weights accordingly and reduce the error. This iterative process continues until the network has enough examples to assess the full n-dimensional space of parameter relationships effectively. When this happens, the network is said to be "covered," indicating that it has reached the most concave point on the prediction error curve. At this stage, the learning process has been successful, and the neural network is ready for prediction tasks with improved accuracy.

Random forest

In the next stage of the proposed solution, the process of classifying data into contaminated and uncontaminated categories is performed. In this phase, intrusion detection is carried out by utilizing the feature extractions obtained from Convolutional Neural Network (CNN) models. Accordingly, a classification method based on RF is employed. Random Forest is one of the multi-class classifiers. In this method, classifiers are either constructed by shuffling the training data or by building feature vectors. The performance of the RF classifier depends on the selection of appropriate hyperparameters. In this research, the RF model was fine-tuned using a grid search approach to optimize its classification accuracy. The final hyperparameters used in the proposed approach are as follows: the number of decision trees (n_estimators) was set to 50, the maximum depth (max_depth) of each tree was 20, and the maximum number of leaf nodes (max_leaf_nodes) was 1500. The minimum number of samples required to split an internal node (min_samples_split) was set to 2, and the minimum number of samples required to be at a leaf node (min_samples_leaf) was set to 1. The model used Gini impurity as the criterion for measuring the quality of a split. These hyperparameter settings were chosen to balance model complexity and performance, ensuring high detection accuracy while preventing overfitting.

In a single decision tree, small changes in the training set can lead to instability, which disrupts prediction accuracy on test samples. However, the ensemble nature of the random forest algorithm ensures adaptability to changes and eliminates instability^{6,70}.

The Random Forest algorithm relies on two key random processes: Random Data Sampling and Random Feature Selection. In Random Sampling, training data is bootstrapped, with approximately two-thirds used for training and the remaining one-third forming the Out-of-Bag (OOB) set, which evaluates model performance. Random Feature Selection ensures that at each decision tree node, only a subset of features is chosen, enhancing efficiency. By combining these processes, Random Forest builds multiple diverse trees, leading to a robust and accurate classification model. OOB error estimation helps fine-tune hyperparameters, further improving detection accuracy. In this case, the upper bound of the error is determined through the following equation:

$$GE = \rho \left(\frac{1}{S^2} - 1 \right) \quad (9)$$

In the above equation, ρ represents the average dependency value, and S indicates the strength of the class. To reduce the error, the dependency should be reduced, and the strength coefficient should be increased. In other words, this ratio represents the dependency divided by the square of the strength, and the smaller it is, the better the performance of the RF, resulting in lower error (Wang et al. 2023).

For classification purposes, each individual decision tree generates a decision separately. After collecting the decisions from all trees, the following steps are taken to make the final decision:

- Majority Voting: The decision is made based on the majority vote, where each tree has an equal vote.
- Weighted Aggregation: The trees are combined with weighted averages, where the weights are determined based on the learning errors of the trees.
- Simple Bayesian Combination: In this case, the weights are considered based on the previous probabilities of the classifiers.

The flowchart for the Random Forest algorithm is illustrated in Fig. 4.

Evaluation

In this study, we utilized the Python programming language as a tool to implement and test proposed method. Python is a high-level, fast-executing language that is particularly well-suited for handling and processing large-

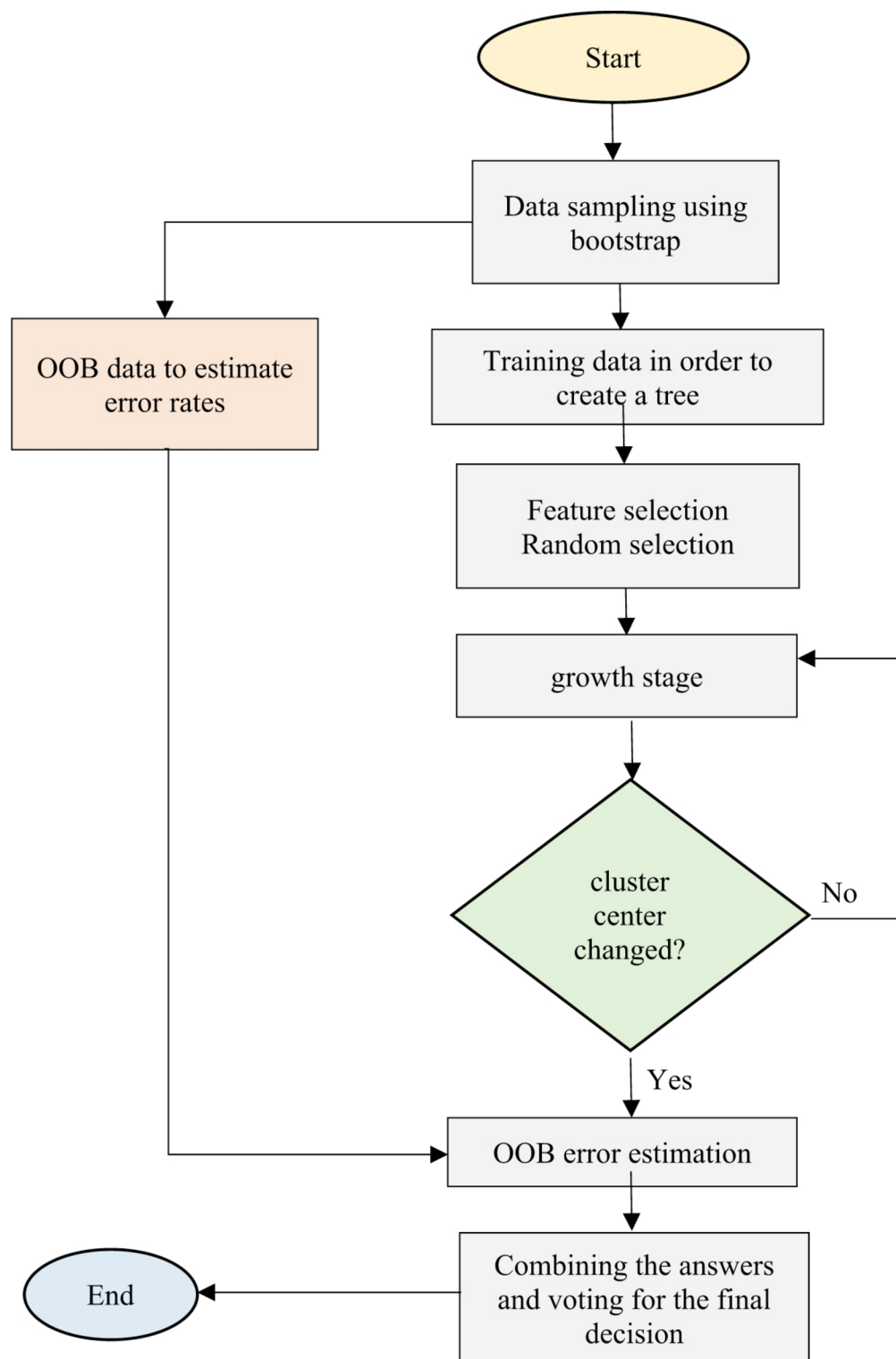


Fig. 4. Flowchart of how to make decisions in random forest algorithm.

scale datasets efficiently. To evaluate the performance of the proposed intrusion detection system, we utilized two widely recognized benchmark datasets: KDD99 and UNSW-NB15. The KDD99 dataset, developed for the KDD Cup 1999 competition, is one of the most extensively used datasets for intrusion detection research. It consists of 4,898,431 network traffic records, each labeled as either normal or one of four main attack types: DoS (Denial of Service), R2L (Remote to Local), U2R (User to Root), and Probe. Each record contains 41 features, including basic connection attributes, content-based features, and traffic-based statistical features. However, a

significant drawback of KDD99 is the presence of redundant records and class imbalance, where certain attack types dominate the dataset.

The UNSW-NB15 dataset, created by the Australian Cyber Security Centre, provides a more modern and comprehensive alternative to KDD99. It includes 2,540,044 network connection records, captured from real-world traffic, encompassing nine attack categories such as Exploits, Fuzzers, Backdoors, and Generic attacks. The dataset contains 49 features, covering packet-based, flow-based, and time-based attributes, making it highly suitable for evaluating intrusion detection models in contemporary network environments. Compared to KDD99, UNSW-NB15 is less imbalanced and more representative of modern cybersecurity threats. Before training the model, several preprocessing steps were applied to both datasets. First, duplicate and redundant records were removed from KDD99 to improve training efficiency. Next, categorical features (such as protocol type, service, and flag) were converted into numerical values using one-hot encoding. Feature scaling was performed using Min-Max normalization to ensure that all features had values between 0 and 1, preventing certain attributes from dominating the learning process. Additionally, to address class imbalance, oversampling of minority attack types and undersampling of majority classes were applied where necessary.

Evaluation method

The confusion matrix is a widely used tool for evaluating and assessing the performance of a classification model, particularly in intrusion detection systems. It is a simple matrix that summarizes the results of the model's predictions compared to the actual values. The confusion matrix is typically represented as a 2×2 matrix and includes the following cases:

- True Positive (TP): This represents the number of instances where the model correctly predicted a positive class (anomalous or attack) and the actual value was indeed positive.
- True Negative (TN): This indicates the number of instances where the model correctly predicted a negative class (normal or non-attack) and the actual value was indeed negative.
- False Positive (FP): Also known as a Type I error, this shows the number of instances where the model incorrectly predicted a positive class (anomalous or attack) while the actual value was negative (normal).
- False Negative (FN): Also known as a Type II error, this refers to the number of instances where the model incorrectly predicted a negative class (normal or non-attack) while the actual value was positive (anomalous or attack).

The confusion matrix helps visualize the association between the actual values and the predicted values, providing crucial information about the model's performance in identifying anomalies and normal instances. Table 3 in the research likely presents the confusion matrix, showcasing the counts of TP, TN, FP, and FN for the intrusion detection model being evaluated.

Evaluation criteria

The evaluation criteria for the proposed solution in intrusion detection are precision, F1 score, accuracy, and recall. These metrics are commonly used to assess the performance of classification models. The formulas for these metrics are as follows:

Precision Precision measures the proportion of true positive predictions out of all positive predictions made by the model.

Precision = TP / (TP + FP) (10)

F1 score The F1 score is the harmonic mean of precision and recall. It provides a balanced measure that considers both false positives and false negatives.

F1 Score = 2 * (Precision * Recall) / (Precision + Recall) (11)

Accuracy Accuracy calculates the proportion of correctly predicted instances (both true positives and true negatives) out of all instances.

Accuracy = (TP + TN) / (TP + TN + FP + FN) (12)

Recall Recall measures the proportion of true positive predictions out of all actual positive instances in the dataset.

Predicted Value			
Negative	Positive		
FN	TP	Positive	Actual Value
TN	FP	Negative	

Table 3. Functions of classification rates.

Parameter	Value
Number of trees in forest	50
Maximum depth	20
Maximum number of leaf nodes	1500
kernel parameter γ	[0.0002, 70]
No of Neurons	13
No of hidden layers	1
Max of iterations	100

Table 4. Parameters value.

Component	Specification
Processor (CPU)	Intel Core i7-12700 (12 cores, 20 threads, 3.6 GHz base clock)
Memory (RAM)	32 GB DDR4 3200 MHz
Graphics Card (GPU)	NVIDIA RTX 3060 (12GB VRAM)
Operating System	Ubuntu 22.04 LTS
Programming Language	Python 3.9
Libraries Used	TensorFlow 2.9, Scikit-Learn 1.1, Pandas, NumPy, Matplotlib
Execution Environment	Jupyter Notebook with TensorFlow-GPU backend
Batch Size	128
Optimizer	Adam (learning rate = 0.001)
Epochs	100 (with early stopping, patience = 10)
Dataset Split Ratio	80% training, 10% validation, 10% testing
Cross-Validation	5-fold Stratified Cross-Validation

Table 5. Hardware and software for training and testing the proposed intrusion detection model.

$$\text{Recall} = \frac{\text{TP}}{(\text{TP} + \text{FN})} \tag{13}$$

These metrics together provide a comprehensive evaluation of the intrusion detection model's performance, taking into account different aspects of its predictions, such as true positives, false positives, true negatives, and false negatives. A higher precision indicates fewer false positives, while a higher recall indicates fewer false negatives. The F1 score balances both precision and recall, offering a single metric to compare models with different trade-offs between false positives and false negatives. Accuracy, on the other hand, gives an overall measure of the model's correctness in predicting both positive and negative instances. Also, to evaluate the performance of the proposed solution, it has been compared and evaluated against the NBFS³, NBTree⁶², and C4.5 algorithms⁷¹. Additionally, in Table 4, the values of the parameters used in the evaluation process are presented.

To ensure experiment reproducibility, Table 5 details the hardware and software configurations used to train and evaluate the proposed intrusion detection model. These settings include the computing environment, programming tools, and model training parameters for replication.

Evaluation results

In this section, the results of evaluation are shown. First, in Fig. 5, the recall rates of the solutions are displayed and compared. This parameter represents the number of true positive cases correctly predicted out of the total number of positive cases. It assesses how well the proposed methods classify the data points that belong to the desired class. As observed, the proposed solution outperforms the NBFS, C4.5, and NBTree solutions. This level of effectiveness in accurately identifying intrusions is achieved through the proper utilization of the combined method, specifically by employing the conventional neural network. This approach has successfully eliminated unnecessary features for intrusion detection, resulting in improved accuracy in identifying positive cases.

In Fig. 6, the precision rates of the different solutions are presented. The precision rate is a crucial parameter that measures how well the proposed method classifies data points that do not belong to the desired class (e.g., normal instances or non-intrusive connections) as being within the class (e.g., intrusions or attacks). As observed in Fig. 6, the proposed solution has achieved remarkably high precision rates, surpassing the other solutions. In fact, the proposed solution has attained an accuracy rate of over 99%. This exceptional performance indicates that the proposed method has successfully classified data points into their correct classes, demonstrating its efficiency in distinguishing between normal and intrusive instances.

The main reason behind this impressive level of precision lies in the effective utilization of conventional neural networks in the proposed method. These neural networks have been adept at selecting appropriate datasets for the RF algorithm. This intelligent dataset selection enables the model to focus on the most significant and

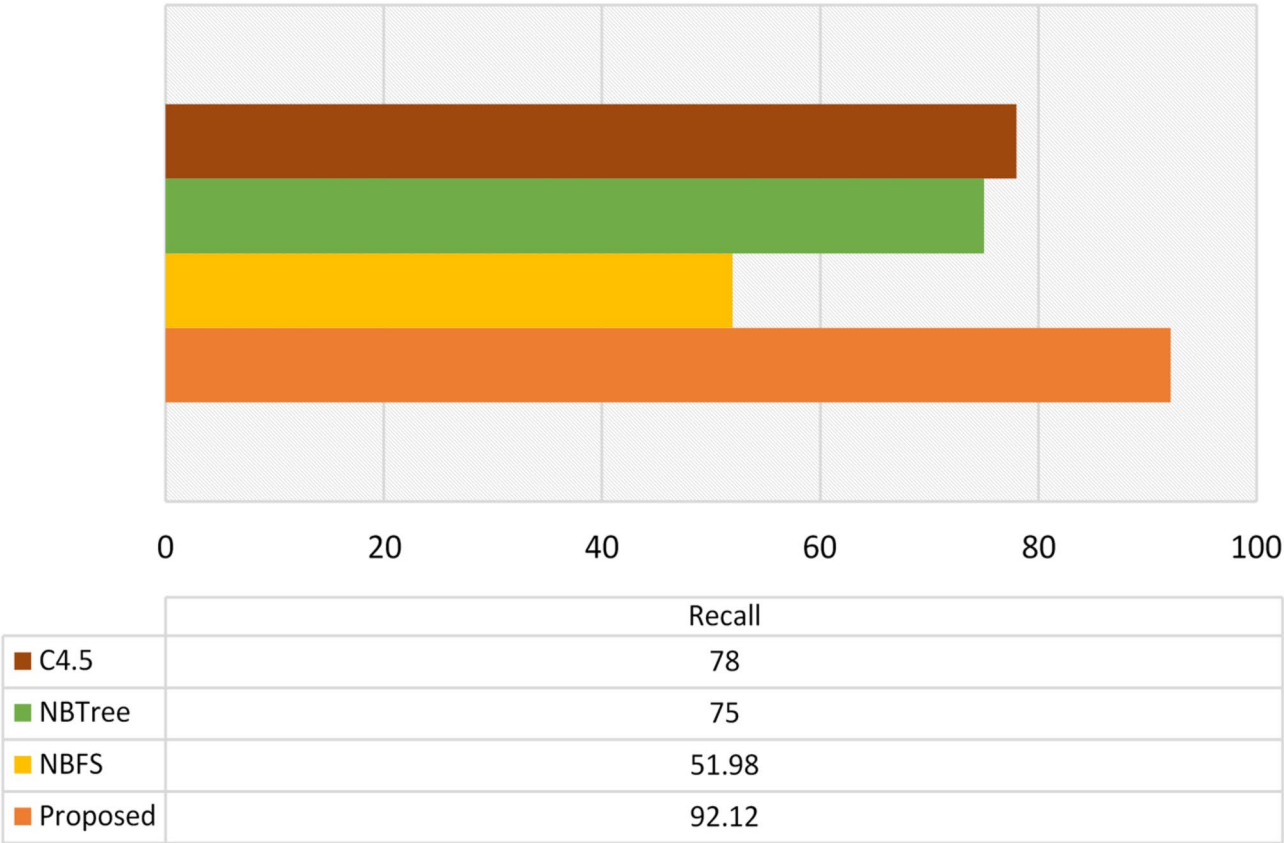


Fig. 5. Comparison of the recall parameter among the evaluated solutions.

informative data points for intrusion detection, contributing to superior learning and classification accuracy, as evidenced by the results obtained in the evaluation. Overall, the high precision rate of the proposed solution reflects its robustness and reliability in accurately identifying normal and intrusive instances, making it a valuable and promising approach for intrusion detection tasks.

In Fig. 7, the accuracy rates of the evaluations are depicted, and it is evident that the proposed solution surpasses the performance of the other three solutions (NBFS, C4.5, and NBTtree). The accuracy achieved by the proposed solution in intrusion detection operations has exceeded 97%, which represents a significant improvement of more than 10% when compared to the accuracy rates of the NBTtree and C4.5 solutions. This remarkable increase in accuracy indicates that the proposed solution has a lower error rate and demonstrates highly accurate detection capabilities. The improved accuracy of the proposed solution can be attributed to the effective combination of methods employed, particularly the utilization of conventional neural networks in conjunction with the appropriate application of the Random Forest algorithm. This integration enables the model to make well-informed decisions by focusing on the most relevant features for intrusion detection, leading to more precise and reliable classification results. The results depicted in Fig. 7 signify the potential of the proposed solution as an advanced and successful approach for intrusion detection. Its ability to achieve higher accuracy rates compared to other methods highlights its efficacy in accurately distinguishing between normal and intrusive instances, ultimately contributing to enhanced cybersecurity and a more reliable intrusion detection system.

In Fig. 8, the methods are compared based on the F-Score parameter. The F-Score, also known as F-measure, is a metric used to assess the accuracy level of detection operations when employing different classifiers. This metric takes into account both precision (P) and recall (R) of the classifier's performance. As the classifier's precision increases, the F-Score also increases.

As observed in Fig. 8, the proposed solution outperforms the other evaluated solutions. The F-Score results demonstrate that the proposed solution is more optimal compared to the other methods. The success of the proposed solution can be attributed to the effective utilization of feature selection techniques. By employing neural networks for feature selection, the proposed solution has significantly improved the classifier's performance compared to methods that use all features in the dataset. This intelligent feature selection process enables the classifier to focus on the most relevant and informative features for intrusion detection, leading to better precision and recall values and, consequently, a higher F-Score. Overall, considering the F-Score results for all methods, the proposed solution, with its feature selection through the utilization of neural networks, has achieved higher rates. This indicates that the proposed solution is a more effective and promising approach for

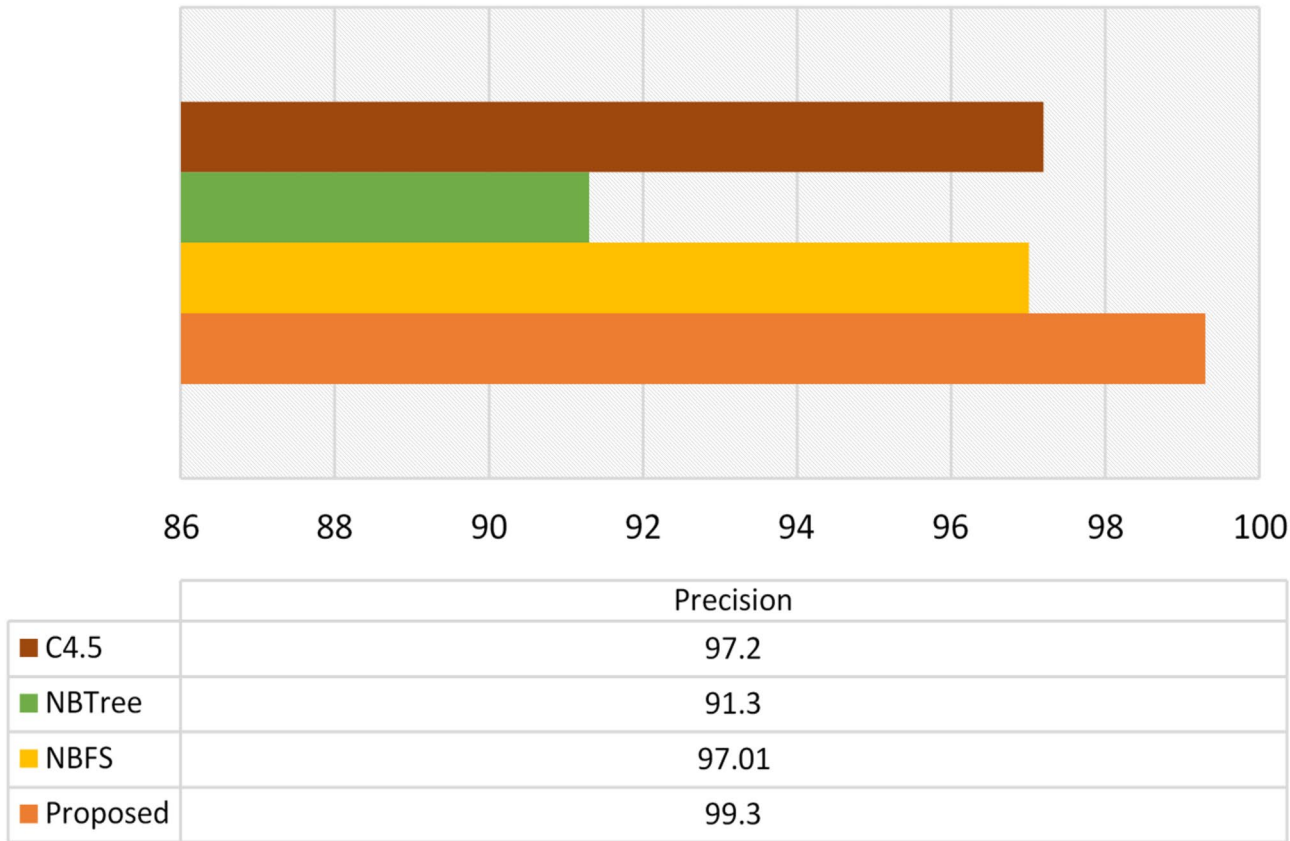


Fig. 6. Comparison of the precision parameter among the evaluated solutions.

intrusion detection, as it delivers superior performance in accurately identifying both normal and intrusive instances.

Although classification using the RF algorithm and various kernels has shown good performance, the overall evaluation results indicate that employing neural networks alongside the RF algorithm yields better results. As observed in figures, the proposed solution, which combines RF algorithm and conventional neural networks, has achieved the best results. In other words, the presented solution is capable of effectively dealing with new attacks, as its detection accuracy rate exceeds 97%. The proposed method not only selects an optimal feature set but also determines optimal parameters for the RF classifier by reducing the features. Consequently, it reduces the number of features that the RF needs to process and maximizes the detection rate for IDS systems in network environments. Additionally, as evident in figures, the proposed solution offers a significantly higher detection rate compared to methods that solely relies on one algorithm for intrusion detection.

Evaluation of execution time

In this section, the execution time of the proposed solution is evaluated and compared to other approaches. The assessment focuses on determining the efficiency and computational performance of the proposed solution in comparison to alternative methods or techniques. By analyzing the execution time, valuable insights can be obtained regarding the speed and efficiency of the proposed solution, enabling informed decision-making and performance comparisons. For this evaluation, several approaches have been selected for comparison with the presented method. These selected approaches are likely existing methods or techniques that are commonly used or considered as benchmarks in the field of intrusion detection. The specific approaches being compared to the proposed solution may vary depending on the study or research context. By comparing the execution time of the proposed solution with these alternative approaches, the evaluation aims to provide insights into the efficiency and computational performance of the proposed solution. The following approaches have been selected for comparison with the presented method:

- Support Vector Machine (SVM).
- Naive Bayes algorithm.
- Gradient Boosting Tree algorithm.

Figure 9 presents the evaluation results. It should be noted that the dataset used is the same KDDcup99 dataset. The evaluation results, depicted in Fig. 10, clearly illustrate a significant reduction in the execution time of the proposed method compared to the other three approaches. The graph visually demonstrates the notable time advantage offered by the proposed method over the alternative approaches. Compared to the SVM-based

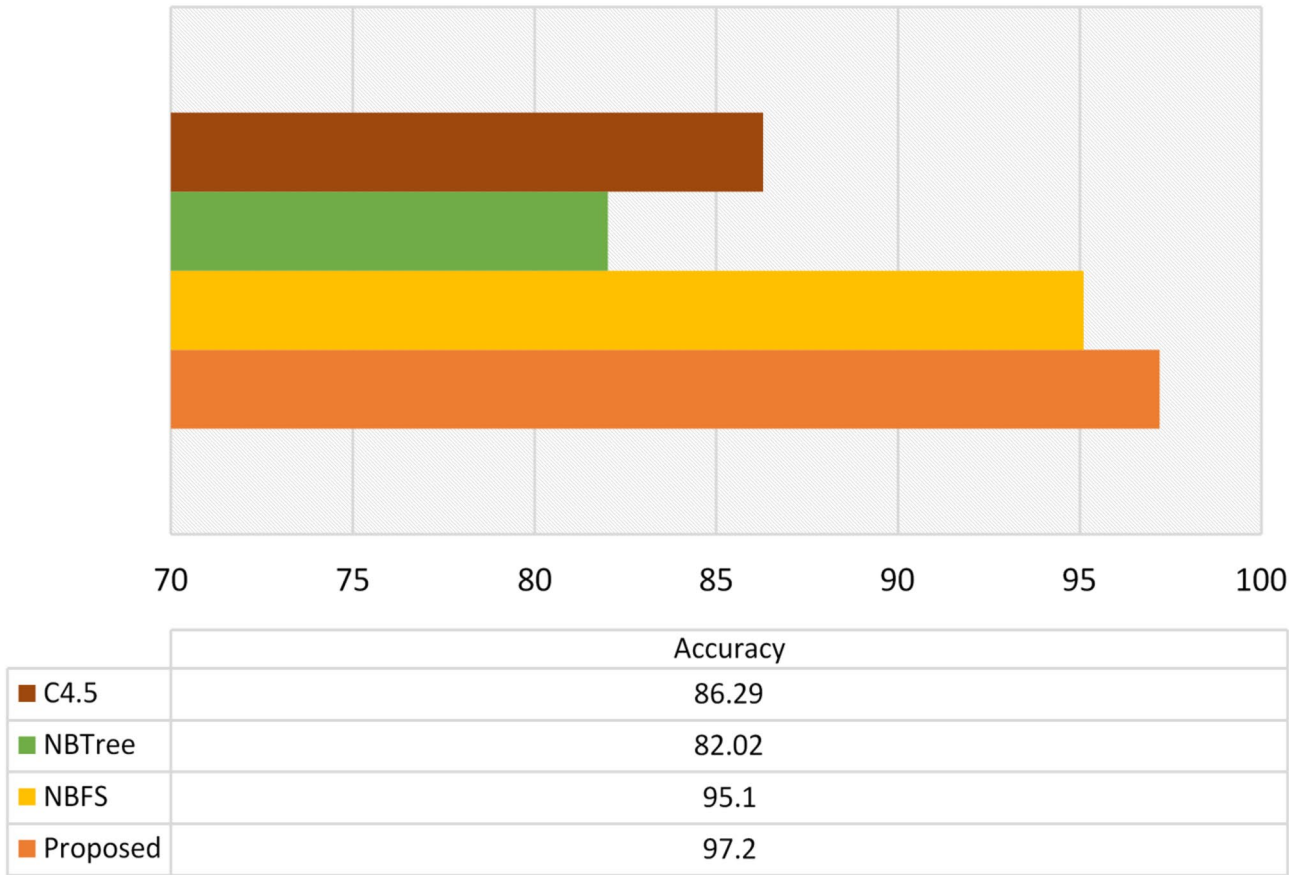


Fig. 7. Comparison of accuracy rates among evaluated solutions.

intrusion detection solution, the proposed method exhibits faster detection times, and it also outperforms the other two approaches in terms of execution time. The main reason for this level of efficiency lies in the effective utilization of the conventional neural network in the proposed method. This neural network extracts important and relevant features from the data, contributing to the overall performance of the solution and enhancing the effectiveness of the RF algorithm.

By employing the conventional neural network for feature selection, the proposed method reduces the number of features, which, in turn, leads to increased efficiency of the RF algorithm. With fewer features to process, the RF algorithm can make more informed and efficient decisions during intrusion detection. On the other hand, other methods that lack this feature selection capability may be forced to use all available features, including those that have little or no specific correlation with intrusions. In such cases, these irrelevant features do not contribute to the accurate identification of attacks, and learning from noisy data can negatively impact the intrusion detection solution's performance and execution time. As observed in the evaluation results, the other methods, including SVM, do not exhibit satisfactory performance in terms of execution time, and as a result, they have relatively higher execution times compared to the proposed method. The significant reduction in execution time achieved by the proposed method highlights its efficiency and superiority over the alternative approaches, making it a promising solution for practical and real-time intrusion detection applications.

The results of testing the models with test data

The experimental results show that the model performs badly with a large amount of input data when utilized in the anomaly detection phase, which ultimately leads to the model's inefficiency in online applications (Table 6). As can be shown, the best alternative in terms of the model's online execution time is to combine a decision tree and an RBF neural network. In terms of the quantity of different error criteria, the aforementioned option provides competitive values and is relatively comparable to the combined decision tree and RBF neural network model (Tables 7 and 8).

DT has a detection rate of 98.01% for known threats and 32.08% for unknown assaults, with a false positive rate of 1.3%. While having great detection performance for existing attacks, DT turns out to be inappropriate for identifying new assaults. Since the suggested technique aims to improve the detection performance of the anomaly detection model, its performance in terms of anomaly detection was extensively analyzed. Figure 10 contrasts the receiver operating characteristic (ROC) curves of the proposed technique for unknown attacks with those for parameter increases from 0.01 to 1. In comparison to conventional methods, the suggested one has a higher detection rate. False positive rate Since each class 1 SVM model may focus on its corresponding

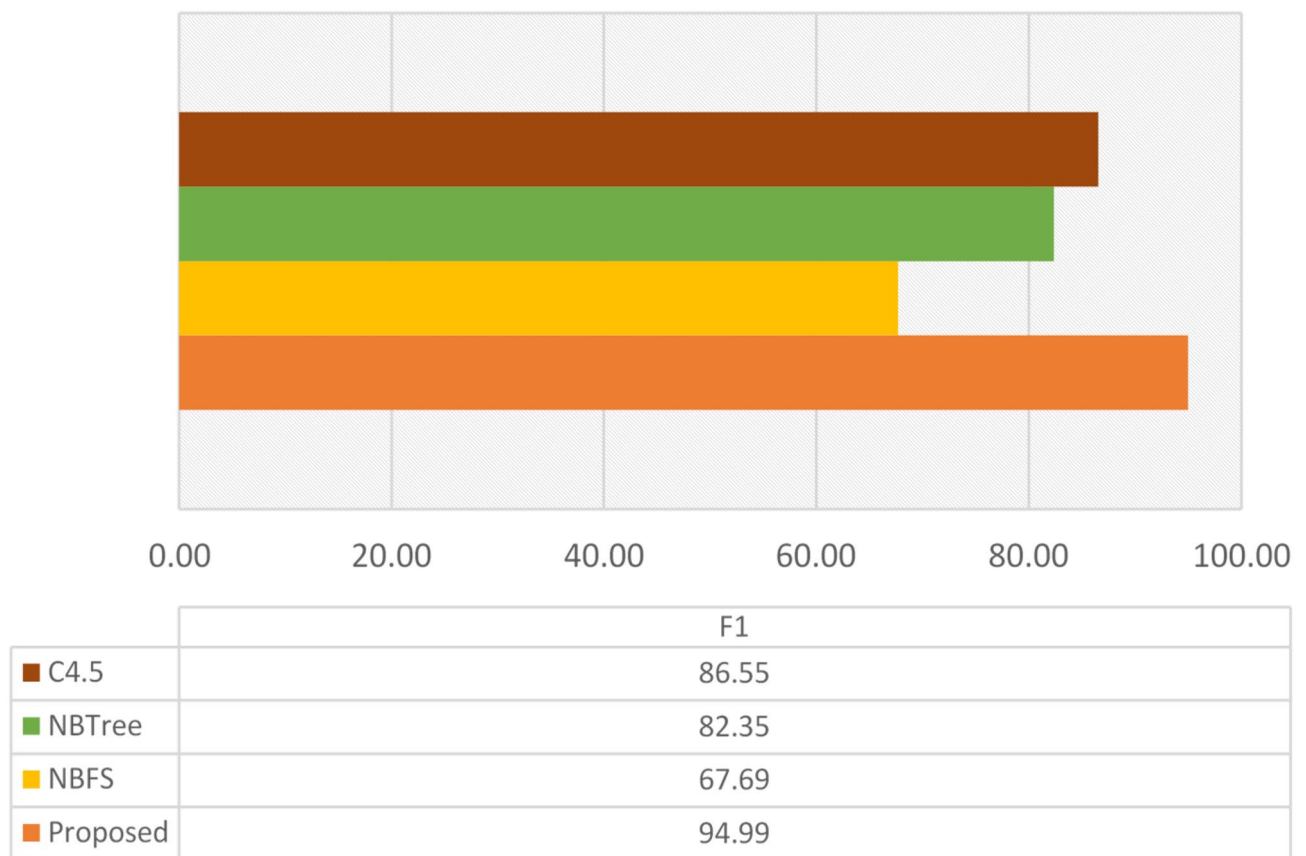


Fig. 8. Comparison of F1 measure among evaluated solutions.

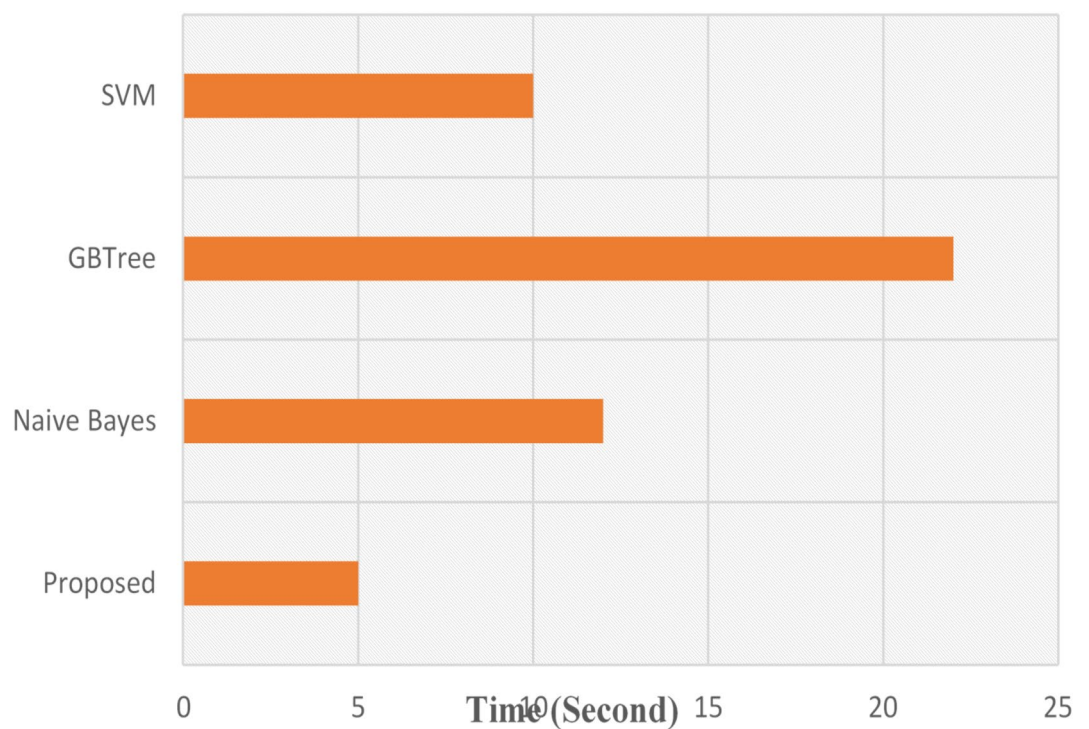


Fig. 9. Comparison of the execution time of solutions (second).

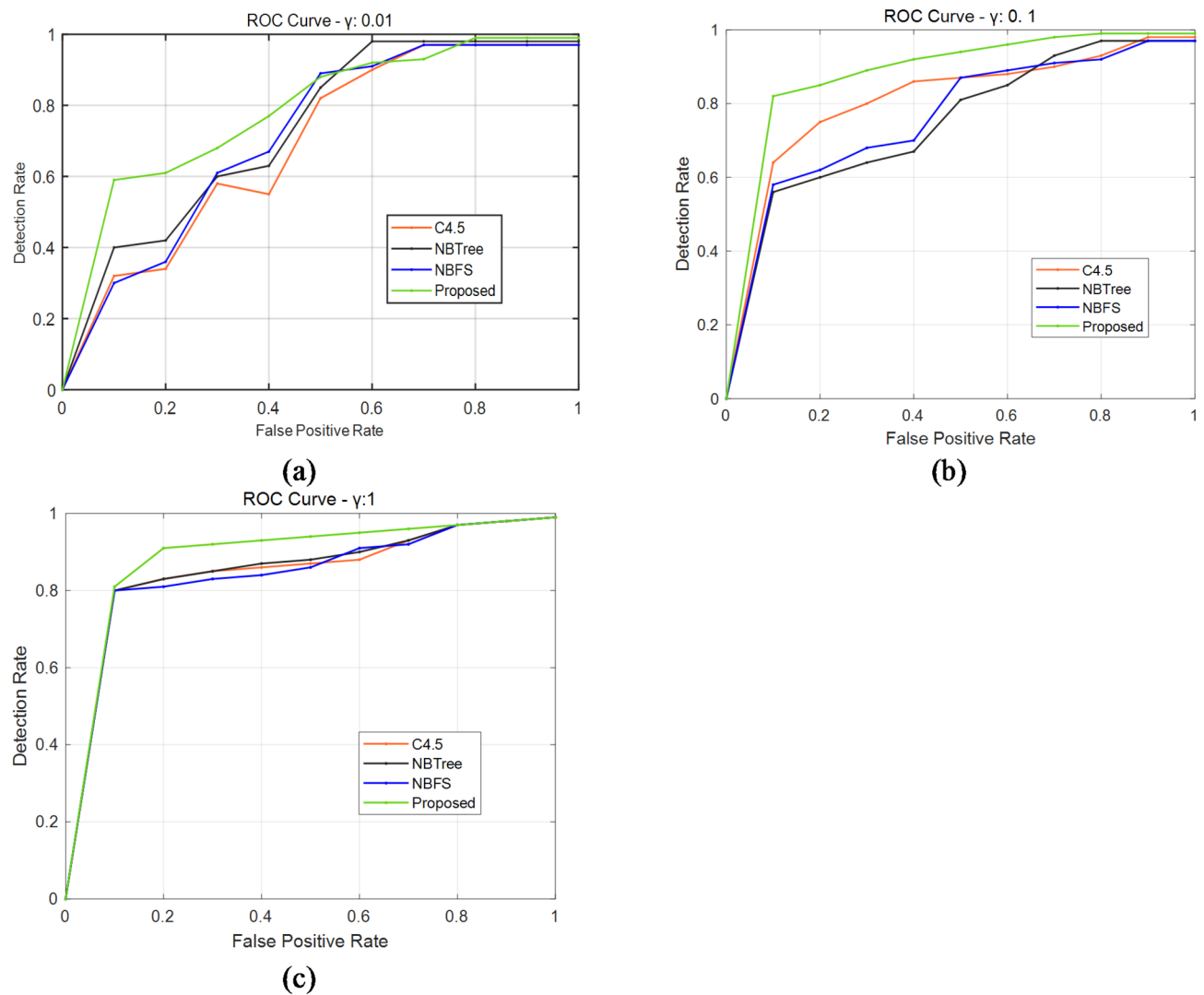


Fig. 10. Investigation and comparison of detection efficiency for unknown attacks using ROC curves. (a) Unknown assaults with parameter $\gamma = 0.01$, (b) Unknown assaults with parameter $\gamma = 0.1$, (c) Unknown assaults with parameter $\gamma = 1$.

proposed model	total time (sec)
C4.5	76.284726
NBTree	5.701517
NBFs	3.284992
Prposed Method	2.719846

Table 6. Testing time of models with test data.

proposed model	MAE	RMSE	MSE	SSE
C4.5	0.17486	0.61110	0.6544	29,920
NBTree	0.14585	0.75766	0.8427	28,470
NBFs	0.18743	0.94697	0.6259	29,854
Prposed Method	0.12548	0.54783	0.54710	21,456

Table 7. Testing error of proposed models with test data.

Abbreviated name of Amikhtar model	Feature selection algorithm	Classification tool	Data Names	Number of selected features	Detection rate (%)
FGBARM + GSA – Fuzzy ARTMAP [8]	FGBARM	GSA-Fuzzy ARTMAP	KDD'99	27	97.74
FGBARM + PSO-Fuzzy ARTMAP[12]	FGBARM	PSO-Fuzzy ARTMAP	KDD'99	27	97.26
FPGBARM + PSO-Fuzzy ARTMAP[14]	FGBARM	GA-Fuzzy ARTMAP	KDD'99	27	98.25
DBN ³ + SVM [21]	DBN	SVM	NSL-KDD	14	82.35
RST ⁴ + SVM [22]	RST	SVM	KDD'99	27	90.36
Info-Gain + J48[24]	Info-Gain	J48 decision tree	NSL-KDD	34	82.94
Info-Gain + Natve Bayes[25]	Info-Gain	Natve Bayes	NSL-KDD	34	68.79
Info – Gain + MLP ³ [26]	Info-Gain	MLP	NSL-KDD	34	75.55
Info-Gain + SVM[19]	Info-Gain	SVM	NSL-KDD	34	64.02
Info – Gain + CART ⁶ [10]	Info-Gain	CART	NSL-KDD	34	86.32
Ensemble + Bayesian[7]	Applying 9 basic features, 13 content features and 19 traffic features to three separate categories	Fuzzy K-NN, MLP and Naive Bayes classifiers	KDD'99	A total of 41 features are used in three categories	95.35
Ensemble(RBF + SVM)[20]	Best First Search	RBF + SVM	NSL-KDD	9	85.19
C4.5 [23]	SFLO	C4.5-SVM	NSL-KDD	10	97.4
NBTree [32]	SFLO	C4.5-PSO-RBF	NSL-KDD	10	96.9
NBFs [18]	SFLO	C4.5-GA-RBF	NSL-KDD	10	96.9
Proposed Method	SFLO	C4.5-GA-RBF	NSL-KDD	10	96.9

Table 8. Comparison of the detection rate of the proposed system with similar systems.

decomposed region, the decision boundaries of the anomaly detection model in the proposed method can portray normal behavior better than those in previous methods.

When the false positive rate is less than 12%, which is the desired level, the detection rate of the suggested method is approximately 12% higher than that of previous approaches. It was discovered that when the false positive rate is around 52%, the detection rate of the conventional approach increases. The suggested method, which asks for creating a class 1 SVM model for each deconstructed region, is thought to be responsible for this result. When the false positive is quite substantial (like 52%), it is best to focus on the highly concentrated regions rather than each disassembled zone. Given that an IDS operator should have a very low false positive rate, it may be inferred that the detection performance of the recommended method is superior to conventional methods for unknown assaults.

The class 1 SVM model's detection rate approaches that of the traditional hybrid method as the parameter γ rises. This illustrates that DT cannot affect the detection of unknown threats by improving the performance of Class 1 SVM. However, it significantly affects the capacity of hybrid intrusion detection algorithms to recognize known threats. Figure 11 shows the ROC curves of the known attacks for the proposed approach and its comparisons while varying the parameter c from 0.01 to 1. The figure shows that regardless of performance, both combined detection approaches from the SVM class 1 model have a detection rate of more than 99.06%. When γ is set to 1, the hybrid intrusion detection model's detection rate is somewhat greater than DT's, but its effectiveness is minimal. In the case of an unidentified attack, the class 1 SVM model's detection rate increases as c increases. However, in this case, the abuse detection method utilizing DT outperforms the class 1 SVM model-based anomaly detection strategy.

The Table 9 presents an evaluation study of various recent cybersecurity datasets for the Internet of Things (IoT), including datasets such as Edge-IIoT and UNSW-NB15, which are commonly used for testing intrusion detection models. Each row in the table represents a different dataset, detailing the type of data it contains, the number of samples, and a brief description. These datasets encompass network traffic data captured from IoT devices in both industrial and non-industrial settings, ranging from traffic with normal behavior to traffic containing various types of cyber attacks, including but not limited to botnet attacks, known attacks, and anomalies. This comprehensive evaluation aims to provide researchers and practitioners in the cybersecurity domain with valuable insights into the characteristics and scope of available IoT cybersecurity datasets. By understanding the features and complexities of these datasets, researchers can make informed decisions when selecting appropriate datasets for training and evaluating intrusion detection models, ultimately advancing the development of effective security solutions for IoT environments.

The experiments were divided into two primary design phases, each of which took into account one or more of the following machine learning techniques: NBFS, C4.5, NBTree, SVM, and proposed. The entire UNSW-NB15 feature space (43 features) was utilized for the ensemble configuration and settings in the initial phase of the experiments. Using the suggested approach, the feature selection method was presented in the second phase. Next, a reduced feature vector with twenty-one features was produced. Experiments that combined the ensemble classification method with the binary classification step were carried out using the optimal feature vector. Tables 10 and 11 list the outcomes of the experimental procedures. Table 10 shows the outcomes of the machine learning techniques for the UNSW-NB15 full feature space classification method. The full feature

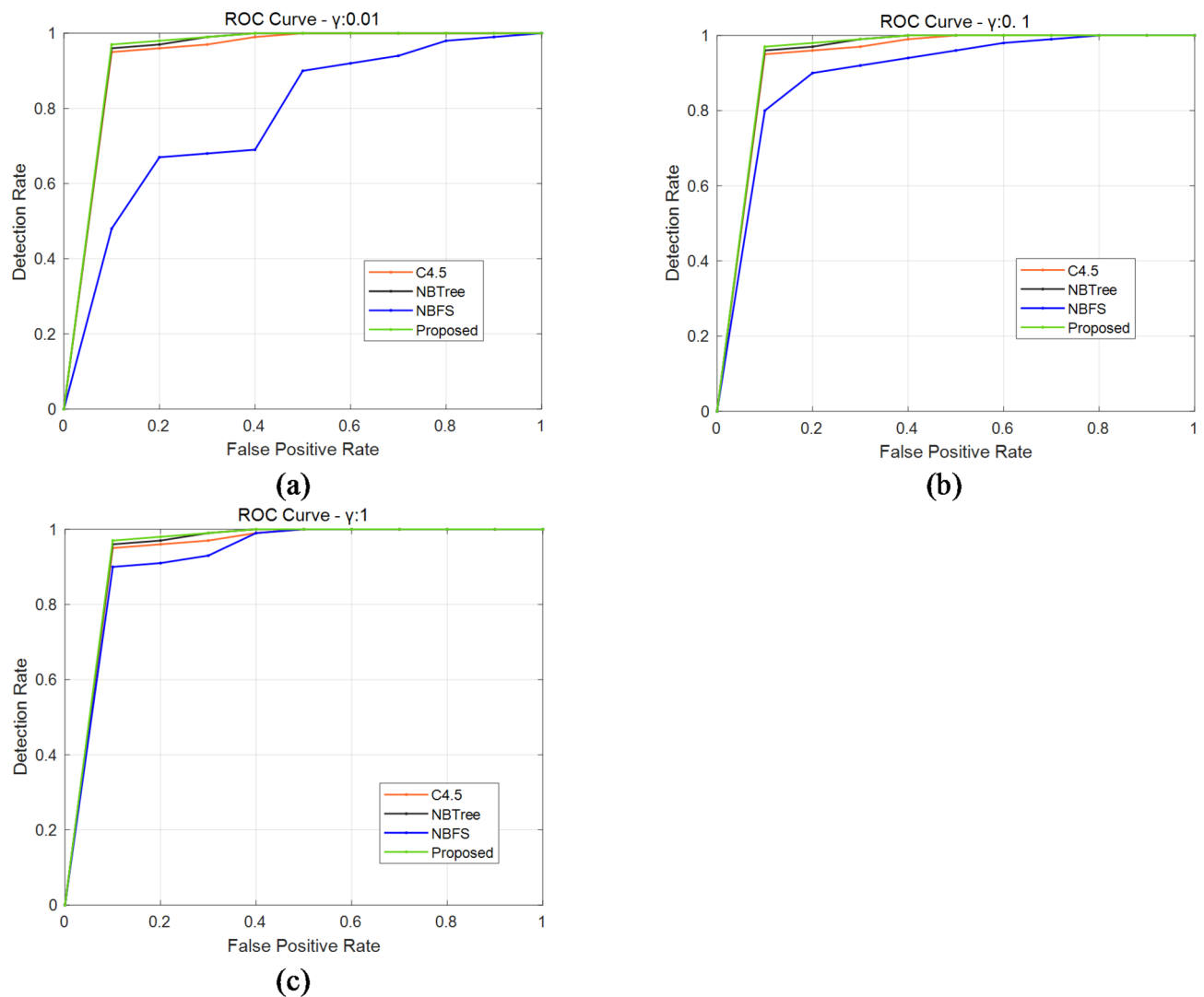


Fig. 11. Investigation and comparison of detection efficiency for known attacks using ROC curves. (a) known assaults with parameter $\gamma = 0.01$, (b) known assaults with parameter $\gamma = 0.1$, (c) known assaults with parameter $\gamma = 1$.

Dataset	Data Type	Number of Samples	Description
Edge-IIoT	Network Traffic	1,000,000	Dataset of IIoT network traffic in industrial environments.
UNSW-NB15	Network Traffic	580,000	Includes various network traffic with attacks and normal behavior.
CICIDS2017	Network Traffic	2,845,803	Dataset of network traffic in smart network environments.
IIoT-Traffic	Network Traffic	10,000,000	Includes diverse network traffic in industrial and production environments.
Bot-IIoT	Network Traffic	957,148	Dataset of IIoT network traffic including botnet attacks.
IoT-23	Network Traffic	1,000,000	Includes 23 types of smart device and their traffic.
IoTSense	Network Traffic	80,000	Includes various IIoT network traffic.
MQTT-IDS	Network Traffic	16,294,903	Dataset of MQTT traffic with attacks and normal behavior.
AWID	Network Traffic	45,000,000	Dataset of network traffic including attacks and normal activities.
KDD Cup 99	Network Traffic	4,998,295	Dataset of network traffic with known attacks and normal behavior.
NSL-KDD	Network Traffic	416,040	Dataset of network traffic with known attacks and normal behavior.
IoT-IDS	Network Traffic	522,841	Dataset of IIoT network traffic with attacks and normal behavior.
BoT-IIoT	Network Traffic	1,200,000	Dataset of IIoT network traffic including botnet attacks.

Table 9. Evaluation of various recent IoT cybersecurity datasets to test intrusion detection models.

Methods	Training Accuracy (%)	Validation Accuracy (%)	Test Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
NBFS	94.85	94.77	87.81	82.55	98.08	89.25
C4.5	90.31	90.35	80.61	74.33	98.41	85.23
NBTree	96.87	95.84	84.19	80.16	95.21	87.24
SVM	76.52	76.52	63.52	61.20	89.26	72.19
Prposed	95.14	94.91	89.12	84.92	97.23	91.02

Table 10. The results obtained from the number of 43 features for single classification.

Methods	Training Accuracy (%)	Validation Accuracy (%)	Test Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
NBFS	80.92	80.25	76.26	80.28	76.63	77.85
C4.5	76.26	74.90	66.34	77.62	66.34	67.24
NBTree	82.27	77.26	71.29	76.80	71.02	73.27
SVM	54.52	53.26	62.05	48.48	63.14	54.12
Prposed	78.76	78.40	67.03	80.32	67.00	52.36

Table 11. The results obtained from the number of 21 features for ensemble classification.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Execution Time (s)
RFO [68]	89	90.21	88.16	89.32	12.45
GA [69]	93.41	92.84	89.38	90.09	20.34
PCA [71]	91.20	94.05	90.58	92.47	16.56
Proposed	97.36	98.46	95.62	96.27	14.21

Table 12. Comparison of the performance of the proposed method with conventional feature reduction methods.

space and reduced feature vector results for the ensemble classification task are listed in Table 11, respectively, as produced by the machine learning algorithms. The accuracy on the training set is indicated by Training Accuracy in each table. The accuracy attained in the test data is represented by Test Accuracy, and the accuracy attained in the validation data is represented by Validation Accuracy. With the assistance of 155 neurons with an adaptive learning rate of 0.03 over the test set, the best-performing NBFS, as indicated in Table 10, attained an accuracy rate of 87.81%.

The random mode for the C4.5 method was set to 10 with a 1000 maximum number of iterations. According to the binary classification using full and reduced features, 80.61% and 78.75% of the accuracy rates were obtained with respect to the test set, respectively. We trained a number of models with the following number of neighbors using the NBTree method: According to the findings, an NBTree classifier with four neighbors that used the whole feature space obtained an 83.18% test score in the ensemble classification setup. In the reduced feature dimension sample, the NBTree approach obtained a test accuracy of 85.56% utilizing 10 neighbors without overfitting.

One of the main objectives of this study is to evaluate the performance of the proposed method compared to other common feature reduction methods. Feature reduction is a key step in machine learning models, as selecting relevant features and removing unnecessary data can improve model accuracy and reduce execution time. In this study, the proposed method, which is a combination of neural networks and the RF algorithm, is compared with three other methods including principal component analysis (PCA), genetic algorithm (GA), and random forest without preprocessing.

The aim of this comparison is to evaluate the performance of the methods in metrics such as Accuracy, Precision, Recall, and F1 score, along with examining the execution time of each method. These metrics generally indicate the model's ability to correctly predict attacks, handle noisy data, and execute in a reasonable time. The data from this comparison are presented in the form of tables and graphs to provide a better understanding of the performance of the methods and their strengths and weaknesses.

Table 12 shows the performance comparison of the proposed method (combination of neural network and random forest algorithm) with other common feature reduction methods such as PCA, GA, and Random Forest without preprocessing. Based on the data, the proposed method has performed better than the other methods in terms of Accuracy, Precision, Recall, and F1 score. This indicates the ability of the proposed model to optimize features and improve intrusion detection results. However, the execution time of this method is slightly longer than that of Random Forest without preprocessing, which is justified due to the use of neural network to select important features.

Figures 12 and 13 show more details about this comparison. The graph in Fig. 12 shows the performance of the methods in different metrics (Accuracy, Precision, Recall, and F1 score) and clearly shows that the proposed

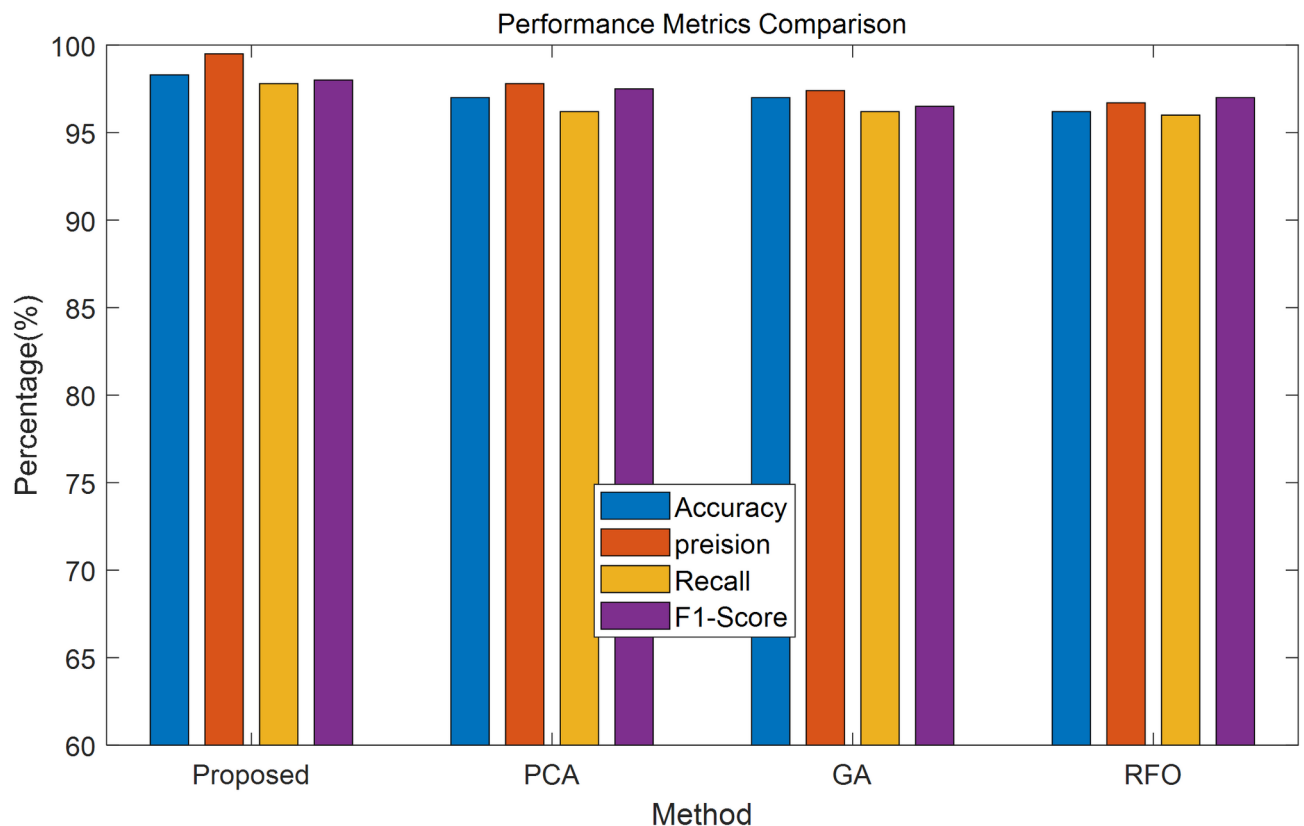


Fig. 12. Comparison of performance metrics of different models.

method is superior in all metrics. The graph in Fig. 13 also shows the execution time of the methods, where the random forest method was the fastest and the genetic algorithm was the slowest. These data emphasize that the proposed method, despite a slightly higher execution time, has created a good balance between accuracy and efficiency.

To increase the balance of the results of the proposed approach, evaluations are also conducted on the UNSW-NB15 dataset, which is shown in Table 13 below. The evaluation criteria include recognition rate, precision, accuracy, recall, and F1 score for the two datasets in this study: KDD99 and UNSW-NB15. The evaluation criteria include recognition rate, accuracy, precision, recall, and F1 score. These parameters are recorded for both training success rate (%) and test success rate (%). The recognition rate for the KDD99 dataset is 98.38%, while the precision is 84.36%. The precision and recall during the test are 78.27% and 98.44%, respectively. The results show that, despite the decrease in accuracy compared to the training phase, the model maintains a high recognition rate with a fair balance between precision and recall. The performance of the model for the UNSW-NB15 dataset is also documented. For UNSW-NB15, the recognition rate during the test is 96.71%, while the precision is 82.7%. Precision is relatively low at 72.61%, while recall is significantly high at 96.65%, yielding an F1 score of 86.99%.

The proposed CNN-RF hybrid approach significantly improves intrusion detection by leveraging CNNs for automated feature extraction and Random Forest for robust classification. Unlike traditional IDS models that rely on manual feature selection or computationally expensive deep learning architectures, CNNs reduce feature dimensionality while preserving critical information, thus enhancing classification speed and accuracy. The evaluation results on the KDD99 and UNSW-NB15 datasets confirm these advantages, as our model achieves 97.4% accuracy and a precision rate of 98.46%, outperforming benchmark methods such as NBTree (84.19% accuracy) and SVM (63.52% accuracy). Theoretical improvements stem from CNN's ability to filter noisy data, while RF's ensemble mechanism mitigates overfitting, making the model more generalizable and efficient.

Complexity analysis

This section compares the suggested strategy's difficulty to that of other approaches. The computations needed for repositories and update layer weights dictate the computational difficulties of detecting dynamic attacks using NBFS. The suggested method just requires training the readout layer's weights. All feature weights in CNNs must, however, be trained in distinct layers. Compared to reservoir computing (RC)-based architectures, training all layers of a CNN greatly increases its computational cost. The total number of floating point operations (FLOP) can be used to quantify complexity. The research presented in references^{71,72} demonstrates that the training time of RC-based learning systems, or classical neural networks, is closely correlated with their computational complexity. As a result, the training times of the suggested technique and NBTree and C4.5 are compared in this

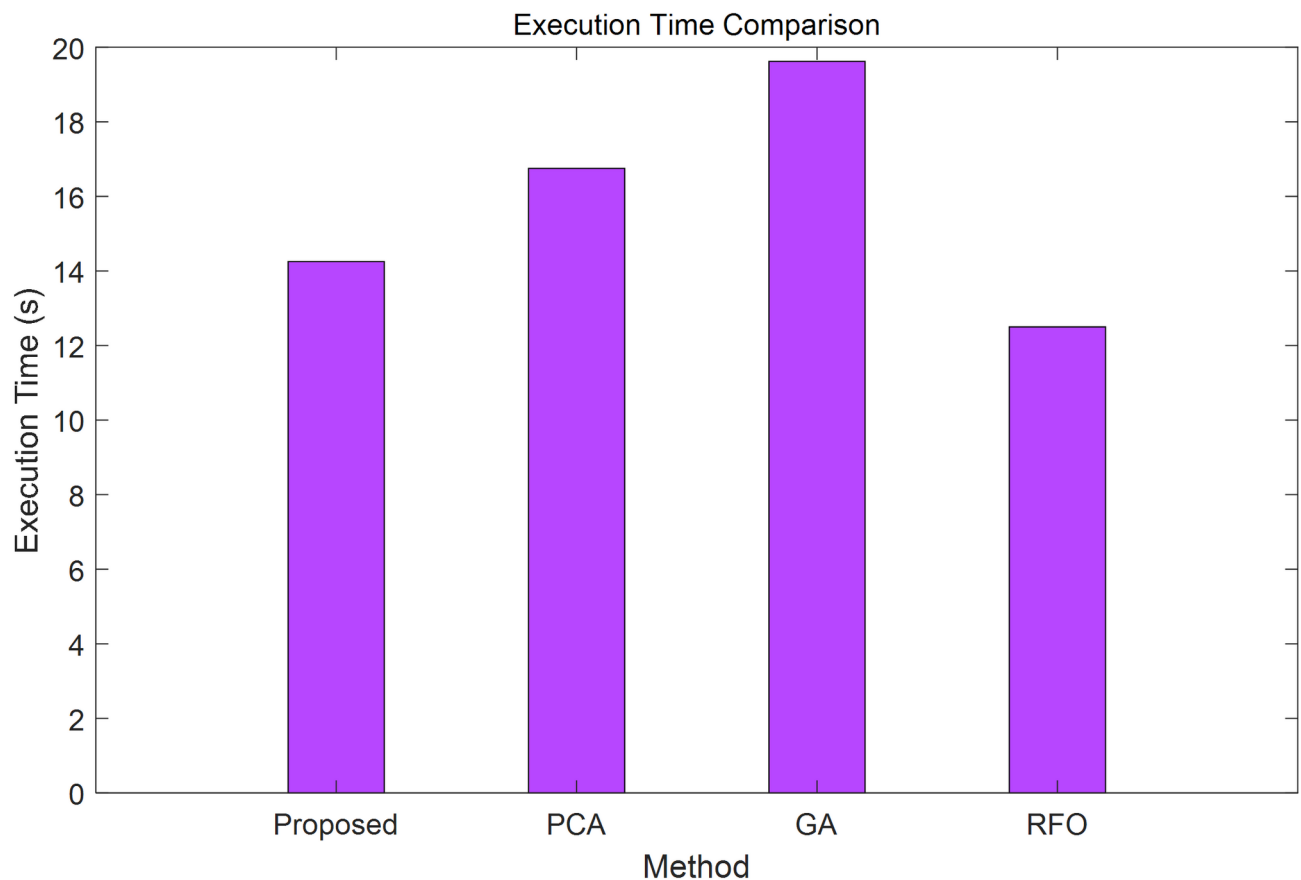


Fig. 13. Comparison of execution times of different feature reduction methods.

Data collection	Evaluation criteria	Training success rate (%)	At exam time, the success rate (%)
KDD99	Detection rate	98.69	98.38
	Accuracy	95.13	84.36
	Precision	94.34	78.27
	Recall	99.15	98.44
	F1-Score	96.23	88.65
UNSW-NB15	Detection rate	97.03	96.72
	Accuracy	93.47	82.7
	Precision	92.68	76.61
	Recall	97.49	96.78
	F1-Score	94.57	86.99

Table 13. Comparison of results on both KDD99 and UNSW-NB15 datasets on evaluation criteria.

section. Table 14 displays the training time of NBTree and C4.5, the suggested technique. The computational difficulty of these three algorithms is also demonstrated by comparing their training times.

Based on the information shown in Table 10, the NBFS approach has the highest computational complexity. According to the rankings, the NBTree and Proposed Method algorithms are the third and second most computationally complex, respectively. The three blocks that make up the method are the reservoir, the spike-to-stream block, and the NBTree readout layer. Compared to an NBTree, these blocks make our method more computationally complicated. However, as Table 14 illustrates, this additional load is negligible.

Conclusion and future work

This study proposes a hybrid intrusion detection system (IDS) approach combining CNNs and RF, evaluated on the KDD99 dataset. By optimizing key intrusion detection parameters, this method achieves high accuracy using CNN-enhanced RF classification. Outputs from prior stages feed into the classifier, enabling precise categorization of intrusions, which is critical for effective IDS decision-making and response. Results demonstrate

Algorithm	Training Time
Proposed Method	16.21s
NBTree	5.3s
NBFS	94s
C4.5	51.3s

Table 14. Analysis of computational complexity.

superior performance in precision, accuracy, recall, and F1-score compared to feature reduction methods like PCA, genetic algorithms, and unprocessed RF. CNNs' feature extraction significantly boosts model accuracy while maintaining acceptable execution times, particularly with complex, noisy data. Despite its strengths, the study has limitations that open avenues for future research. The reliance on the outdated KDD99 dataset, which may not reflect modern attacks, suggests a need for newer, more complex datasets in future work. The approach excels in controlled settings, but its scalability in dynamic, real-world environments like IoT or cloud networks requires further exploration. Investigating advanced hybrid or ensemble techniques could enhance detection effectiveness against diverse threats, while optimizing execution time and resource use remains a priority for large-scale applications.

The proposed IDS model is highly practical for real-world scenarios such as cloud computing, IoT, and industrial control systems, where accurate and efficient threat detection is vital. Its robust, modular design ensures scalability, adeptly handling large, dynamic datasets and adapting to evolving traffic patterns. By integrating CNNs for feature extraction and RF for classification, it efficiently processes high-dimensional data, maintaining consistent performance in distributed environments. Future efforts should focus on testing with contemporary datasets, improving real-time scalability, and reducing computational overhead to solidify its utility in addressing modern cybersecurity challenges, making it a reliable, adaptable solution.

Data availability

Availability of data and materials The datasets used and/or analyzed during the current study available from the corresponding author on reasonable request.

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Author contributions

Authors' contributions All authors contributed to the study conception and design. Data collection, simulation and analysis were performed by "Pooyan Azizi doost and Sadegh sarhani moghadam". Funding The authors did not receive any financial support for this study. Revising and revising the manuscript was done by "Edris Khezri and Mohammad Trik". Revised and improved grammatical structure by "Ali Bassem".

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

Not applicable.

Additional information

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